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资本市场自由化是否推动数字化转型？来自中国 A 股纳入 MSCI 新兴市场指数的证据

Does Capital Market Liberalization Drive Digital Transformation? Evidence from Chinese A-Shares' Inclusion in the MSCI Emerging Markets Index

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摘要

新兴经济体资本市场自由化水平不足不利于企业融资，进而制约其转型发展。本文以中国 A 股分阶段纳入 MSCI 新兴市场指数为外生制度冲击，基于 2010-2022 年间 2,712 家上市公司数据，借助双重差分识别策略，并结合事件研究和倾向得分匹配等方法，研究新兴市场经济体的资本市场自由化能否推动企业数字化转型。

实证结果表明，纳入 MSCI 能够显著促进企业数字化转型提高 0.39 个对数点，相当于处理前水平提升约 48%。动态事件研究结果显示，纳入 MSCI 前的企业数字化转型不存在系统性趋势差异，纳入 MSCI 后的促进效应持续且具有累积特征。上述结论在安慰剂检验、分期处理诊断、替代聚类标准误、受限样本检验和替换被解释变量的检验中均成立。异质性分析发现，资本市场自由化对企业数字化转型的促进效应在高技术行业和经济较发达地区更为显著，表明吸收能力和制度互补性会影响自由化收益。机制检验显示，资本市场自由化通过缓解融资约束、强化外部监督与信息披露促进企业数字化转型，知识溢出机制未通过实证支持。

本文研究结论表明，国际金融一体化可作为新兴市场企业数字化升级的重要催化机制，因此新兴经济体应持续推进金融体制改革，着力促进企业数字化转型。

关键词：资本市场自由化；数字化转型；MSCI 纳入；中国 A 股；新兴市场；计算文本分析；双重差分

Abstract

The insufficient level of capital market liberalization in emerging economies hinders corporate financing, thus restricting their transformation and development. This paper takes China's phased inclusion in the MSCI Emerging Markets Index as an exogenous institutional shock. Based on data from 2,712 listed companies between 2010 and 2022, and using a difference-in-differences (DID) identification strategy, along with event studies and propensity score matching, this study investigates whether capital market liberalization in emerging market economies can promote corporate digital transformation.

Empirical results show that inclusion in MSCI significantly promotes digital transformation, increasing it by 0.39 log points, equivalent to an approximately 48% increase relative to pre-treatment levels. Dynamic event study results indicate that before MSCI inclusion, there were no systematic trend differences in corporate digital transformation, while the promoting effect after MSCI inclusion is persistent and cumulative. The robustness of these conclusions holds across placebo tests, phase-treatment diagnostics, alternative clustering standard errors, restricted sample tests, and changes in the dependent variable. Heterogeneity analysis shows that the positive effect of capital market liberalization on corporate digital transformation is more significant in high-tech industries and more economically developed regions, indicating that absorptive capacity and institutional complementarity influence the liberalization benefits. Mechanism tests reveal that capital market liberalization promotes corporate digital transformation by alleviating financing constraints and enhancing external supervision and information disclosure, while knowledge spillover mechanisms were not empirically supported.

The conclusions of this study suggest that international financial integration can serve as an important catalytic mechanism for the digital upgrading of firms in emerging markets. Therefore, emerging economies should continue advancing financial system reforms to facilitate corporate digital transformation.

Keywords: Capital market liberalization; Digital transformation; MSCI inclusion; Chinese A-shares; Emerging markets; Computational text analysis; Difference-in-differences

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List of Abbreviations

Abbreviation	Full Term
AI	Artificial Intelligence
ATT	Average Treatment Effect on the Treated
C1	Basic Manufacturing (CSRC Subcategory Code)
C2	Equipment Manufacturing (CSRC Subcategory Code)
C3	High-End Manufacturing (CSRC Subcategory Code)
C27	Transportation Equipment Industry (CSRC Industry Code)
C38	Electrical Equipment Industry (CSRC Industry Code)
C39	Instrumentation Industry (CSRC Industry Code)
CF	Cash Flow
CNINFO	China Securities Information Disclosure Platform
CNRDS	China Research Data Service
CNY	Chinese Yuan
CSMAR	China Stock Market and Accounting Research Database
CSRC	China Securities Regulatory Commission
DID	Difference-in-Differences
DOI	Digital Object Identifier
DT	Digital Transformation
DT_Comp	Composite Digital Transformation Index
EM	Emerging Markets
ERP	Enterprise Resource Planning
FE	Fixed Effects
GTA	GTA Information Technology Co.
H1	Hypothesis 1
H2	Hypothesis 2
H3	Hypothesis 3
HHI	Herfindahl-Hirschman Index
HMM	Hidden Markov Model
IaaS	Infrastructure as a Service
ICT	Information and Communication Technology
IMF	International Monetary Fund

Abbreviation	Full Term
IoT	Internet of Things
IP	Intellectual Property
IT	Information Technology
KZ Index	Kaplan-Zingales Financing Constraint Index
MSCI	Morgan Stanley Capital International
NLP	Natural Language Processing
NPV	Net Present Value
PaaS	Platform as a Service
PSM	Propensity Score Matching
PSM-DID	Propensity Score Matching Difference-in-Differences
QFII	Qualified Foreign Institutional Investor
R2	Coefficient of Determination
RMB	Renminbi
ROA	Return on Assets
ROE	Return on Equity
RQFII	RMB Qualified Foreign Institutional Investor
SaaS	Software as a Service
SD	Standard Deviation
SE	Standard Error
SIC	Standard Industrial Classification
SME	Small and Medium-Sized Enterprise
STAR	Science and Technology Innovation Board
TWFE	Two-Way Fixed Effects

Chapter 1 Introduction

This chapter introduces the central research problem, explains its academic and policy relevance, and defines the thesis scope and objectives. It places the study at the intersection of capital market liberalization and firm-level digital transformation, clarifies why MSCI A-share inclusion provides a credible quasi-experimental setting, and previews the empirical and methodological contributions developed in later chapters.

Beyond basic orientation, the chapter also states the analytical position used throughout the thesis: digital transformation is modeled as an investment process shaped by both internal managerial capability and external capital-market institutions. This perspective shifts the question from *whether* firms transform to *how* financial opening changes the conditions that make transformation feasible, credible, and persistent.

1.1 Research Background

The relationship between capital market liberalization and corporate digital transformation has become a central issue in development economics, corporate finance, and technology management. Over the last two decades, many emerging economies have gradually opened domestic capital markets to international investors. In parallel, firms have faced growing pressure to digitize production, operations, governance, and customer interfaces. Yet the two streams of research are still often studied in isolation. Liberalization studies usually focus on valuation, liquidity, and investment efficiency, while digital-transformation studies emphasize internal capabilities, leadership, and technology readiness[1]. The key question at their intersection remains open: can international capital-market integration raise firm-level digital transformation, and through which channels?

This thesis starts from a boundary question: digital transformation is not only a technological or managerial process, but also a financing and governance process shaped by market institutions. In many emerging markets, digital programs require high fixed costs, involve uncertain returns, unfold over long horizons, and depend on organizational learning complementarities[2]. Even when expected social gains are high,

these features can create financing frictions that delay or discourage adoption[3]. Capital market liberalization may shift these investment conditions by broadening the investor base, lowering financing costs, and strengthening governance discipline[4]. Understanding this link is therefore important for both scholarship and policy.

The Chinese background is especially important for this research. China features scale, diversity, and institutional growth in ways that make it ideal for empirical study[5]. It is one of the world's largest stock markets and a leading hub for industrial digital transformation, yet capital account liberalization has mostly been gradual and driven by policy[6]. This slow process results in notable policy events with predictable timing. Among these, the inclusion of Chinese A-share firms in the MSCI Emerging Markets Index stands out as a key externally imposed reform that increased firms' exposure to international investors.

The practical motivation is equally strong. Policymakers increasingly treat digital transformation as a core source of productivity growth and competitiveness. However, firms differ substantially in their capacity to implement large-scale transformation programs[7]. If access to global capital and investor networks shapes adoption, market-opening reforms may generate broader technological spillovers than is usually assumed[8]. At the same time, liberalization may widen regional and firm-level capability gaps if benefits accrue mainly to firms that already hold structural advantages. This tension makes careful empirical evaluation necessary.

Measurement is another key motivation. Much of the existing evidence relies on managerial surveys or narrow proxies such as IT spending and patent counts. These indicators often suffer from inconsistent coverage, self-reporting bias, and weak comparability across time[9]. To address this issue, the thesis constructs a firm-year digital-transformation index from annual-report disclosures using computational text analysis. Combining this high-coverage measurement strategy with a quasi-experimental design allows the study to provide evidence that is both identification-oriented and broad in empirical scope.

Chen et al.[10] show that increased analyst coverage and improved information accuracy are two mechanisms through which digital transformation strengthens firms'

information environments. Cheng and Masron[11] further argue that regulatory uncertainty can intensify firms' digital responses, highlighting the central role of external institutional conditions in shaping transformation behavior. Du and Wang[12] find that the innovation benefits of digital transformation are significantly amplified by financial support, particularly through capital markets, with effects significant at the 1% level. Zhang and Wang[13] report that digitally transformed firms face lower equity financing costs, whereas Xue and Zhang[14] document improvements in debt-financing outcomes and overall firm performance. Taken together, this evidence suggests that the financial consequences of digital transformation are conditional on financing environment, disclosure quality, and governance structure.

This study builds on prior work while moving beyond correlation-based inference. Earlier studies often focus on one channel or one outcome and do not fully address treatment endogeneity. By contrast, this thesis combines mechanism-oriented analysis, a long panel (2010–2022), and a quasi-experimental policy shock to test whether liberalization is a causal driver of digital transformation intensity. Conceptually, digital transformation is treated as cumulative strategic investment: firms adjust governance, reassess financing constraints, and update expectations over time. Liberalization is expected to influence this path through at least three channels: financing-constraint relaxation, governance and disclosure upgrading through international oversight, and expanded access to cross-border knowledge networks. These channels anchor the empirical framework.

As a result, there are three main motivations. First, the thesis addresses a significant open question about the relationship between firm-level technological upgrading and global finance. Second, it provides methodologically rigorous evidence from a major emerging economy. Third, it offers policy-relevant insights for firms and regulators seeking to align capital-market reform with digital-development objectives.

1.2 Research Questions and Objectives

In light of the aforementioned motivation, this thesis addresses three relevant research problems.

- First, does inclusion in the MSCI Emerging Markets Index, a proxy for capital market liberalization, significantly increase firm-level digital transformation intensity? This is the thesis's primary causal question. Specifically, the study evaluates whether treated firms exhibit a sustained post-inclusion increase in digital transformation relative to otherwise comparable untreated firms.
- Second, do treatment effects vary by technology domain, industry, region, and firm characteristics? Digital transformation is not a single technology, but a bundle of investments with distinct capital-intensity and capability requirements. The thesis tests whether effects are stronger in domains such as AI, big data, and cloud computing; whether responses differ across technology-intensive and traditional industries; and whether regional institutional quality shapes outcomes.
- Third, how does liberalization affect digital transformation? The thesis investigates three channels using theory and prior evidence: funding restrictions, governance and monitoring, and knowledge spillovers. In empirical implementation, this thesis directly tests the first two channels and treats the knowledge channel as theory-grounded but exploratory in the current sample.

Methodologically, the thesis aims to integrate validated NLP-based measurement with quasi-experimental econometrics in a transparent and reproducible large-panel framework. Substantively, it evaluates whether international financial integration can serve as a catalyst for firm-level technological upgrading. From a policy perspective, it provides evidence relevant to regulators, institutional investors, and corporate decision-makers seeking to understand the technology consequences of market-opening reforms.

The thesis follows five operational aims to achieve these objectives. It generates high-coverage firm-year digital transformation metrics from annual reports. It employs baseline and dynamic DID techniques to estimate treatment effects. It evaluates inferential stability through robustness and sensitivity tests. It uses heterogeneous effects to identify boundary conditions. It conducts mechanism evaluations to assess financing, governance, and knowledge channels.

1.3 The Empirical Setting: China's A-Share Market and MSCI Index Inclusion

The empirical background for this argument is the gradual inclusion of Chinese A-shares in the MSCI Emerging Markets Index, a watershed moment in China's financial opening process[15]. Prior to 2018, global investors had limited access to China's domestic equity market due to ownership restrictions, quota systems, and market-access frictions. Although conduits such as QFII, RQFII, and the Stock Connect programs existed, broad passive global allocation to mainland A-shares remained limited[16].

MSCI inclusion changed this environment by connecting eligible A-share companies to benchmark-driven international capital flows[17]. MSCI indexes are often used by global institutional investors and index-tracking funds. Being included in the MSCI Emerging Markets Index affects active portfolio building, analyst focus, and investor relations practices, in addition to passive investment flows[6]. The reform therefore modified firms' exposure to external financing and oversight conditions in a way that is economically significant.

The phased inclusion process is particularly useful for identification. Phase I was implemented in June 2018 with 234 large-cap A-shares at a 5% inclusion factor following the June 2017 announcement. By November 2019, Phase II expanded coverage to 472 securities with a higher inclusion factor. This phased implementation provides clear treatment timing and observable pre- and post-policy windows. A key advantage of this setting is that inclusion criteria were largely rule-based and externally determined. Eligibility was based on market capitalization, liquidity, and Stock Connect accessibility rather than on discretionary judgments about firms' digital strategies[18]. Although no real-world setting is fully random, this institutional design strengthens the plausibility that inclusion captures exogenous variation in international capital access relative to firms' underlying digital transformation trajectories.

China is also an ideal context because firms operate under substantial heterogeneity in regional development, industry composition, ownership structures, and governance quality. This heterogeneity allows the thesis to test whether liberalization effects are

uniform or conditional on absorptive capacity[19]. Zhang et al.[20] show that treatment effects vary systematically with regional market development, ownership structure, and firm size, with stronger responses in certain non-Eastern and non-state-firm contexts. Embedding these moderation patterns within the MSCI setting helps move beyond average treatment effects toward a richer understanding of distributional outcomes.

1.4 Overview of Research Design and Data

The empirical design combines causal identification with large-scale computational measurement. The baseline model is a two-way fixed-effects DID that compares firms included in MSCI (treated) with firms not included (control) before and after inclusion. This baseline is complemented by event-study estimation for dynamic effects and pre-trend validation, propensity-score-matched DID for improved overlap on observables, and placebo-based falsification tests.

The data cover 35,264 firm-year observations from 2010 to 2022, representing 2,712 Chinese listed firms after sample cleaning. Core financial and governance variables are drawn from CSMAR, while treatment status is derived from MSCI constituent lists. Digital transformation outcomes are constructed from annual-report text using NLP-based keyword extraction across six technology domains: AI/machine learning, big data, cloud computing, IoT, blockchain/distributed ledger technologies, and enterprise digitization programs.

The baseline outcome is a composite index using log transformation of aggregated keyword frequencies. This approach addresses right skewness and enhances interpretability when disclosure dispersion is high. For robustness, alternative outcomes are also used, including domain-specific indices, normalized measures adjusted for report length, and additional transformations to assess sensitivity to scaling choices.

Mechanism analysis is embedded in the design rather than added as a descriptive task. Proxies related to financing constraints and governance quality are estimated through structured regressions to check whether channel patterns match the conceptual framework. The knowledge-spillover channel is retained as a theory-grounded mechanism and interpreted using exploratory evidence in the current sample.

Overall, the design is intentionally triangulated: one institutional shock, multiple estimators, multiple outcome definitions, and multiple mechanism checks. This triangulation increases confidence that the results reflect real economic effects rather than artifacts of any single model choice.

1.5 Significance of Research

This study is significant for three strands of literature: capital market liberalization, digital transformation economics, and computational empirical methods.

- First, it expands the liberalization literature beyond traditional market results. Previous research often focuses on liquidity, valuation, return co-movement, or financing costs. This study demonstrates that liberalization can affect firms' technological paths, thus connecting financial integration to long-term productivity-related behavior.
- Second, it advances the digital transformation literature by highlighting external financial environments as a key factor in shaping corporate digital strategy. While much existing research focuses on internal capabilities, managerial cognition, and organizational change, this thesis demonstrates that access to international capital and investor oversight can significantly influence the speed and direction of transformation.
- Third, it offers mechanism-level evidence that directly tests financing and governance channels under a common treatment design, while treating knowledge spillovers as a theory-grounded exploratory channel with suggestive interpretation.
- Fourth, the thesis enhances methodology by combining quasi-experimental causal inference with validated Chinese-language text analytics at scale.
- Fifth, it contributes to the emerging-market comparative literature by offering high-resolution evidence from China.
- Sixth, it provides policy implications based on micro-level evidence. Regulators interested in the broader development impact of capital market reform can use these findings to assess whether market-opening policies promote strategic upgrading.

1.6 Thesis Structure

The thesis is organized in seven chapters, with each chapter designed to build logically on the previous one.

- **Chapter 1** introduces the research problem, motivation, questions, objectives, empirical setting, and expected contributions.
- **Chapter 2** develops the literature review. It synthesizes foundational and recent studies on liberalization, digital transformation, mechanism channels, and text-based measurement.
- **Chapter 3** presents institutional background and formal hypothesis development. It details China's pre-liberalization capital market structure, the MSCI inclusion process, and the quasi-experimental properties of the reform.
- **Chapter 4** introduces Research Design I focused on causal identification. It describes sample construction, variable definitions, treatment assignment, baseline and dynamic model specifications, and descriptive statistics.
- **Chapter 5** introduces Research Design II focused on computational measurement. It documents corpus construction, Chinese text segmentation, dictionary design, index construction, and validity diagnostics.
- **Chapter 6** reports empirical results and discussion. It presents baseline effects, event-study dynamics, robustness checks, heterogeneity analyses, and mechanism evidence.
- **Chapter 7** concludes with a synthesis of findings, original contributions, limitations, and future research directions.

Chapter 2 Literature Review

This chapter develops the theoretical and empirical foundation of the thesis by integrating research on capital market liberalization, digital transformation, and computational measurement. It reviews the main channels through which international capital access can influence firm behavior, evaluates contextual differences in existing evidence, and identifies unresolved gaps in causal identification and mechanism integration.

2.1 Capital Market Liberalization: Theory and Evidence

2.1.1 Theoretical Foundations of Financial Liberalization

Capital market liberalization generally refers to removing barriers that separate domestic financial systems from international portfolio flows[21]. In theory, opening equity markets can improve allocative efficiency by channeling capital toward more productive firms, reducing financing costs through broader investor participation, and improving risk sharing through more diversified ownership structures[22]. The standard welfare argument predicts stronger gains for firms facing tighter financing constraints[23]. In segmented markets, limited domestic savings, home bias, and information frictions can keep external finance expensive. Liberalization can reduce these distortions by intensifying investor competition and deepening market liquidity[24].

Theoretical work also highlights important non-linearities. The benefits of liberalization depend on institutional quality, legal enforcement, disclosure standards, and market infrastructure[25]. Where institutions are weak, liberalization may amplify volatility or redirect resources toward short-horizon opportunities rather than long-horizon productive investment[26]. At the firm level, liberalization changes the optimization environment facing managers and controlling shareholders[11]: it affects the cost of external finance, investor horizons, and the intensity of external monitoring[27].

The MSCI inclusion mechanism is particularly informative because it combines signaling and flow effects. Inclusion can automatically trigger passive fund allocations while simultaneously signaling market quality and investability to active global investors.

2.1.2 Effects on Corporate Investment, Innovation, and Governance

Empirical studies generally find that market opening influences corporate behavior through investment and governance channels[28]. On the investment side, reduced financing frictions can broaden firms' feasible project options[29]. Firms may increase R&D, capital expenditure, and strategic intangible investments when expected financing constraints lessen[23]. On the governance side, foreign institutional investors can strengthen monitoring and raise transparency expectations.

Evidence from prior work supports this logic. Chen et al.[10] show that digital transformation is associated with higher analyst coverage and improved information accuracy. Du and Wang[12] provide direct evidence that financial support strengthens the innovation effects of digital transformation, particularly in environments with tighter conventional financing constraints. Chen and Zhang[3] and Feng et al.[30] extend this argument to firms' financing structures.

2.1.3 Evidence from Emerging Market Contexts

Emerging markets provide valuable evidence because they combine high growth potential with institutional frictions that highlight financing channels. Studies in these markets often detect stronger real effects of financial reforms compared to more developed markets where constraints are already less severe[31]. The Chinese evidence is especially useful due to detailed data and specific policy information[32]. However, a direct causal link from liberalization shocks to firm-level digital transformation outcomes is less developed.

Zhang et al.[20] show that contextual heterogeneity is central: estimated effects vary with regional market development, ownership structure, and firm size. This pattern implies that pooled average treatment estimates may mask economically meaningful distributional differences. Stiglitz[33] similarly emphasizes that policy shocks interact with organizational absorptive capacity.

2.2 Digital Transformation: Conceptual Framework and Determinants

2.2.1 Defining Digital Transformation: Technological and Organizational Dimensions

Digital transformation is more than technology acquisition. It is an ongoing, firm-wide reconfiguration of value creation, operations, and organizational coordination through digital technologies[34]. A narrow input-based view focused only on hardware and software spending misses the strategic and organizational conditions that determine whether digital investment translates into productivity gains. Conceptually, the process has at least three layers[14,35]: infrastructure adoption, process transformation, and organizational transformation.

These layers depend on each other. Infrastructure without process integration can yield limited results[36]. Process changes without organizational incentives can prevent progress. Organizational commitment without funding can remain superficial[14]. Therefore, digital transformation is best viewed as a coordinated investment journey rather than a single technological event[2].

2.2.2 Core Technology Domains: AI, Big Data, Cloud Computing, IoT, and Blockchain

The technology areas highlighted in this thesis align with key trends in modern corporate digital transformation. Artificial intelligence and machine learning support prediction, optimization, and automation across both front-end and back-end operations[37]. Big data systems enable companies to turn large, fast-moving information into operational insights. Cloud computing offers scalable infrastructure that reduces fixed ICT costs[38]. Internet of Things technologies link physical assets to digital monitoring and control systems[39]. Blockchain and distributed ledger technology provide traceability, tamper resistance, and coordination across organizational boundaries[40]. Enterprise digitization initiatives combine these technologies with ERP upgrades, process engineering, and platform strategies.

Empirically, these domains are related but not interchangeable margins of transformation. AI/ML and advanced analytics are usually data- and talent-intensive, with higher setup costs and longer learning curves. Cloud adoption often lowers entry barriers and accelerates deployment, while IoT initiatives require coordinated redesign of software and physical production systems. Blockchain applications are often narrower, but can be highly relevant in traceability-intensive supply chains. As a result, measured digital intensity can move at different speeds across domains even under similar financing conditions[41-42].

This heterogeneity implies that aggregate digital indices should be interpreted as averages over technologies with different risk profiles, adjustment speeds, and implementation bottlenecks, rather than as a single homogeneous investment decision. That interpretation is important for later empirical sections, where treatment effects are estimated on both composite and technology-specific outcomes.

2.2.3 Drivers, Barriers, and Firm-Level Heterogeneity

Drivers of digital transformation include financial slack, managerial capability, data infrastructure, and complementary human capital. Recent studies also emphasize organizational execution quality: cross-functional coordination, digital leadership commitment, and internal data-governance standards all increase the probability that projects scale beyond pilot stages[43].

Barriers are equally important. High uncertainty about payoffs can delay investment, especially for firms under pressure to meet short-horizon performance targets. Talent shortages in data engineering, AI deployment, and cybersecurity create bottlenecks that raise implementation risk[43]. Legacy IT architectures and fragmented databases further increase integration costs and extend rollout cycles.

These frictions imply heterogeneous response capacities. Larger firms may capture stronger scale economies in platform investment and digital talent recruitment[44]. Firms in economically developed regions may access deeper supplier ecosystems and larger pools of specialized labor. By contrast, resource-constrained firms may prioritize modular adoption and postpone system-wide upgrades.

Ownership and market structure also matter. Non-state firms often face tighter product-market discipline, which can strengthen incentives to adopt productivity-enhancing digital strategies[45]. Overall, the literature indicates that average effects can conceal substantial cross-sectional dispersion, motivating explicit heterogeneity analysis by region, industry, and firm characteristics in the empirical chapter.

2.2.4 Digital Transformation in Emerging Markets: Opportunities and Constraints

Emerging markets offer a digital transformation landscape with both growth opportunities and structural constraints[46]. In these settings, access to external capital markets can play a catalytic role. International investors may provide not only financing, but also governance pressure, benchmarking standards, and stronger emphasis on long-horizon capability building[47].

At the same time, external financing does not mechanically translate into successful digital upgrading. The pass-through from capital access to implementation depends on absorptive capacity, including managerial quality, organizational readiness, and local complementary infrastructure. This mechanism helps explain why treatment effects differ materially across firms and regions in emerging economies.

China is especially informative because advanced digital ecosystems coexist with substantial regional and firm-level inequality in capabilities and financing conditions[48]. This combination provides a strong context for identifying not only whether liberalization affects digital transformation, but also where and for whom the effect is strongest.

2.3 Mechanisms Linking Capital Market Liberalization to Digital Transformation

2.3.1 The Financing Constraint Channel

The financing-constraint channel is the most direct link between liberalization and digital transformation. Digital programs often require substantial upfront spending on software systems, data architecture, cloud migration, machine-learning platforms, and organizational training[49]. By broadening access to equity investors and lowering re-

quired returns, liberalization can reduce the user cost of capital for these investments[50].

Existing studies provide direct support for this channel. Hordofa[51] shows that capital-market financial support significantly strengthens the innovation effects of digital transformation, while Liu[15] documents lower equity financing costs among firms undergoing digital upgrading. Together, these findings suggest that financing and digital transformation are mutually reinforcing. The channel is unlikely to be homogeneous across firms: firms with initially binding financing constraints should exhibit larger responses[52].

2.3.2 Corporate Governance and International Investor Monitoring

A second mechanism is governance upgrading through international investor monitoring. Foreign institutional investors often demand higher disclosure quality, stronger board accountability, and clearer strategic communication[27]. Under these conditions, managers face stronger incentives to prioritize long-term value-enhancing projects and reduce agency-driven underinvestment.

Ma et al.[53] provide supporting evidence that digital transformation is associated with improved information environments through increased analyst coverage. Li[54] further shows that governance effects can interact with ownership structure.

2.3.3 Knowledge Spillovers and Global Best Practice Diffusion

A third mechanism operates through information and knowledge spillovers. Capital market liberalization boosts interactions between domestic firms and global investors, analysts, and intermediaries[8,22]. Through investor meetings, analyst reports, benchmarking exercises, and governance engagement, firms can access implementation templates and performance standards[55]. This mechanism aligns with broader themes in the literature[24,56-57]. Evidence from competitive manufacturing environments further suggests that digital-technology spillovers can be strategically state-dependent across rival firms rather than mechanically uniform[58].

2.3.4 Market Signaling, Certification, and Reputational Effects

Liberalization can affect digital transformation through signaling and certification processes. Inclusion in a major global benchmark is often interpreted as recognition of minimum standards in accessibility, liquidity, and market relevance[54]. This signal can lower external uncertainty about firm quality. Dong[19] further distinguishes substantive transformation from symbolic signaling, an important distinction when interpreting post-inclusion disclosure behavior.

2.4 Text-Based Measurement of Firm Characteristics

2.4.1 Computational Approaches in Corporate Finance and Economics

The rise of text-as-data methods has revolutionized empirical corporate research by enabling high-frequency, large-sample analysis[59]. In the era of digital transformation, computational text methods are especially valuable because direct accounting measures are often incomplete[60]. Methodologically, text-based approaches enhance scalability and consistency compared to purely manual coding[61]. However, computational methods need careful validation[62].

2.4.2 NLP-Based Measurement of Innovation and Digital Activity

Afzal[63] shows that Chinese-language annual reports can provide credible NLP-based indicators of digital activity. Kumar[64] further argues for domain-cluster keyword extraction rather than generic buzzword counts. Li et al.[65] position NLP-based measurement as a complement to, rather than a substitute for, financial indicators.

2.4.3 Validation Strategies for Text-Derived Indices

Validation is central to any text-based empirical strategy. This thesis applies three validation layers: content validity, construct validity, and robustness validity. Content validity is established through careful dictionary design and expert review. Construct validity is evaluated by comparing text-derived indices with related observable variables. Robustness validity is assessed through alternative outcome definitions and algorithmic

specifications.

2.4.4 Challenges in Chinese-Language Disclosure Environments

Chinese-language textual analysis presents technical and interpretive challenges[9,65]. Chinese text segmentation is complex because characters are written without spaces. Another challenge is variation in strategic disclosure[59]. A third challenge involves distinguishing between intention and implementation.

Despite these challenges, Chinese annual reports remain some of the most comprehensive and standardized data sources for long-term firm-level analysis. With careful design and validation, they offer a strong empirical foundation for studying digital transformation at scale.

2.5 Research Gaps and Positioning of This Study

The literature review points to four core gaps that motivate this thesis.

- **Linkage gap**: Research on capital market liberalization and research on digital transformation have mostly developed independently.
- **Identification gap**: Many studies rely on cross-sectional correlations or panel models that may face endogeneity issues.
- **Mechanism integration gap**: Existing evidence suggests financing, governance, and knowledge channels are all plausible but often examined in isolation.
- **Measurement scalability gap**: Direct digital investment metrics are often incomplete in emerging-market accounting systems.

This thesis addresses these gaps by treating MSCI A-share inclusion as a near-natural experiment to identify the causal effect of liberalization on digital transformation. The reviewed evidence supports this positioning, but prior work rarely integrates these insights within a single coherent design. This thesis is structured to provide that integrated causal framework.

Chapter 3 Institutional Background and Hypothesis Development

This chapter establishes the institutional context and formal hypothesis structure for the empirical analysis. It first explains the pre-liberalization structure of China's A-share market and the role of Stock Connect as implementation infrastructure for foreign participation. It then details the MSCI inclusion timeline and rule-based eligibility criteria, clarifying why the reform generates plausibly exogenous variation in international capital access. Finally, it translates the conceptual framework from Chapter 2 into explicit, testable hypotheses.

3.1 China's Capital Market Structure Prior to Liberalization

3.1.1 The A-Share Market and Domestic Investor Segmentation

China's A-share market—the renminbi-denominated equity market on the Shanghai and Shenzhen exchanges—was historically governed by strict foreign ownership restrictions and capital controls, creating near-complete segmentation from international capital flows[66]. Before liberalization, foreign investors could access mainland equities mainly through QFII (Qualified Foreign Institutional Investor) and RQFII channels, both constrained by quotas and administrative barriers. Under this segmented regime, listed firms relied primarily on domestic capital, typically associated with shorter investment horizons, weaker governance pressure, and limited exposure to global best practices.

Figure 3-1 provides a data-based characterization of this segmented pre-liberalization structure using only pre-treatment observations (2010–2017). The chart reports standardized mean gaps between firms that are eventually included in MSCI (2018–2022) and firms never included. The gaps are economically large and statistically precise for firm size (1.20 standard deviations, $p < 0.01$), total market value (0.80 standard deviations, $p < 0.01$), and independent-director representation (0.47 standard deviations, $p < 0.01$), showing that the subset of firms later connected to global capital was already structurally distinct within the domestic A-share universe.

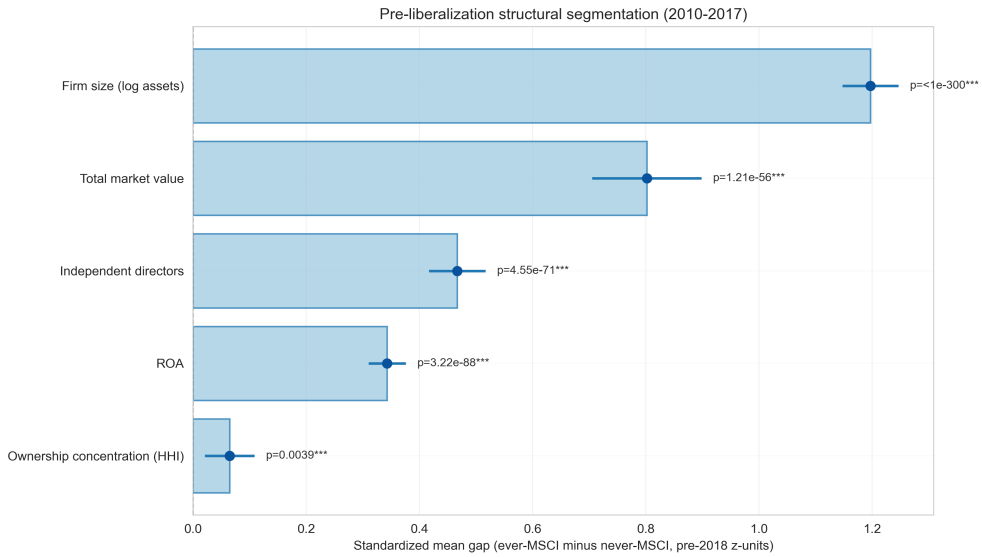

Figure 3-1 Data Profile: Pre-Liberalization Structural Segmentation (2010–2017)

3.1.2 Stock Connect Programs as the Technical Infrastructure for Liberalization

The technical foundation for liberalization was built through Stock Connect: Shanghai–Hong Kong Stock Connect (November 2014) and Shenzhen–Hong Kong Stock Connect (December 2016)[67]. These programs established quota systems, currency-conversion arrangements, and settlement protocols that allowed foreign participation in A-shares while preserving capital-control constraints. By 2018, the platform had matured enough to accommodate the larger portfolio reallocations associated with MSCI inclusion[23].

3.2 The MSCI Emerging Markets Index and the A-Share Inclusion Process

3.2.1 MSCI as a Global Benchmark: Scale, Influence, and Index Composition

The MSCI Emerging Markets Index is a primary benchmark for global institutional investment in developing-market equities. In 2018, roughly $1.9 trillion in assets tracked or benchmarked MSCI emerging-market indices. Because passive funds mechanically follow index composition and active managers often benchmark against it, index membership materially affects capital allocation toward included firms[17].

3.2.2 Inclusion Criteria: Market Capitalization, Liquidity, and Stock Connect Accessibility

MSCI's inclusion criteria were mechanical rather than discretionary: (i) a minimum free-float adjusted market capitalization threshold; (ii) minimum liquidity requirements based on annualized trading volume and turnover ratios; and (iii) accessibility through Stock Connect programs. Importantly, these criteria are independent of firms' underlying digital transformation paths or performance, offering a plausible source of exogenous variation in international capital access.

To make this point data-based, Figure 3-2 estimates a 2017 cross-sectional logit model where the dependent variable is eventual MSCI inclusion. The odds-ratio estimates show strong and statistically significant associations for each rule-based eligibility proxy: log total market value (OR = 17.04, $p < 0.01$), log free-float market value (OR = 2.55, $p < 0.01$), and liquidity proxy (OR = 3.90, $p < 0.01$). This pattern supports the interpretation that inclusion was driven by measurable market-scale and tradability conditions rather than discretionary selection on digital strategy.

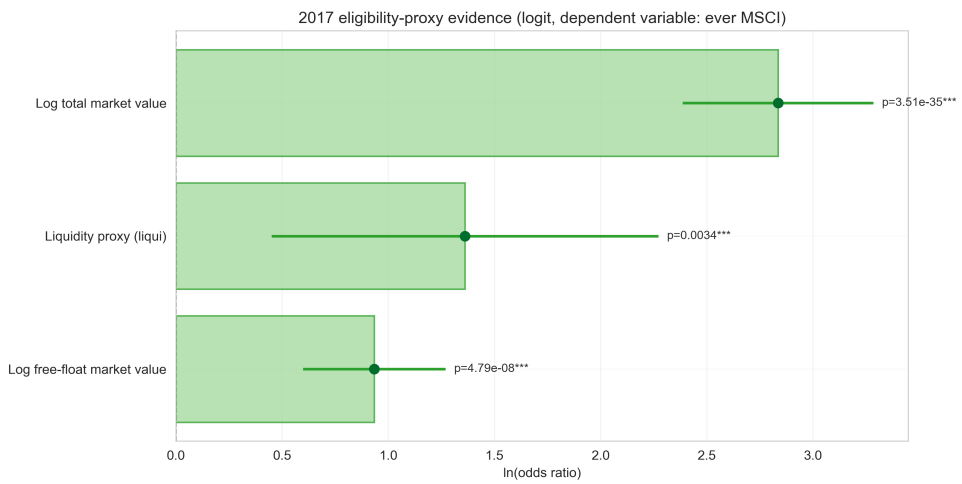

Figure 3-2 Data Profile: Eligibility-Proxy Evidence from 2017 Logit Model

3.2.3 Pre-Announcement and Phased Implementation Timeline

MSCI announced the inclusion decision on June 20, 2017—a full year before the first implementation tranche—providing firms with time to prepare for international investor engagement. The announcement itself constitutes a clean event that allows testing

of announcement effects separately from implementation effects[68].

3.2.4 Phase I (June 2018): 234 Large-Cap A-Shares at 5% Inclusion Factor

The initial tranche, launched on June 1, 2018, incorporated 234 large-cap A-shares at a 5% inclusion factor. This accounted for about 0.73% of the total MSCI Emerging Markets Index. The modest size aimed to evaluate market preparedness and reduce impact, while allowing the included companies to experience international passive fund inflows for the first time.

3.2.5 Phase II (November 2019): Expansion to 472 Securities at 20% Factor

By November 2019, the index grew to include 472 A-shares across large, mid, and small-cap categories, with a 20% inclusion rate. This significantly boosted the presence of Chinese stocks in the index and broadened the scope to include smaller firms, expanding the universe of treated firms.

Figure 3-3 visualizes the realized treatment rollout in the matched panel used for empirical analysis. The number of MSCI firms jumps from zero before 2018 to 151 in 2018 and 374 in 2019, then stabilizes around 372–373 during 2020–2022. In share terms, MSCI firms increase from 4.8% of listed firms in 2018 to 11.4% in 2019 (95% confidence interval approximately [10.3%, 12.5%]), consistent with a sharp, policy-timed treatment expansion.

3.2.6 Quasi-Experimental Properties of the Inclusion Design

MSCI inclusion has three features that make it a strong quasi-experimental setting. First, eligibility criteria are mechanical and not based on firms' digital-transformation strategies. Second, the rollout timeline was set externally by MSCI rather than by Chinese regulators or individual firms. Third, phased implementation creates clear pre- and post-treatment windows with multiple cohorts, which supports staggered-adoption analysis.

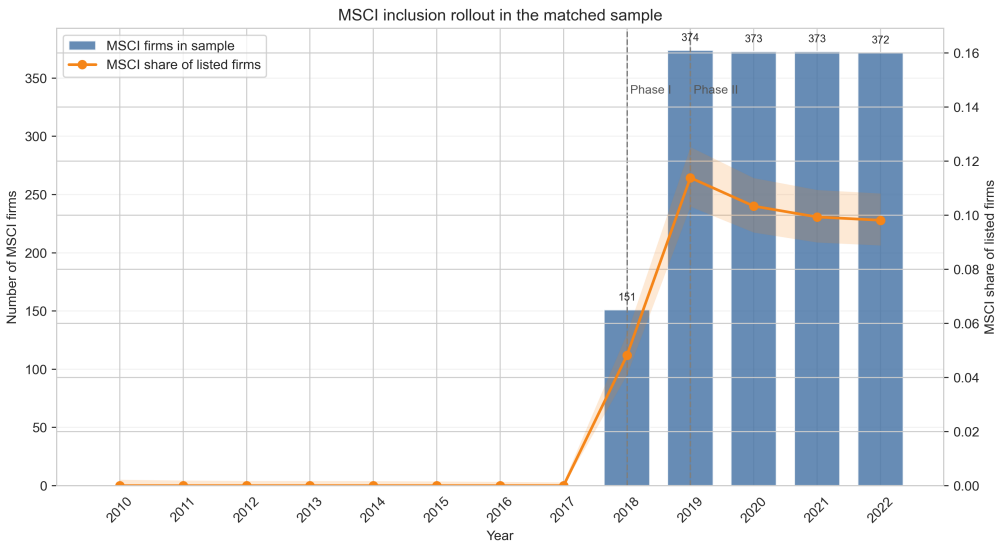

Figure 3-3 Data Profile: MSCI Inclusion Rollout in the Matched Sample

3.3 Hypothesis Development

Building on the institutional evidence above, this section develops hypotheses in three parts: direct effect (3.3.1), transmission mechanisms (3.3.2), and heterogeneity (3.3.3). Consistent with the recommended chapter logic, Sections 3.1 and 3.2 first provide data-based profiles of market structure and inclusion design before formal hypothesis statements.

For clarity, we define the treatment shock as $X_{it} = \text{MSCI}_i \times \text{Post}_t$ and the outcome as $Y_{it} = \text{DT}_{it}$ (firm-level digital transformation intensity). The core causal question is how X affects Y, and through which channels this effect is transmitted.

To formalize channel and moderation logic in advance, let M_{it}^k denote channel-specific mediators ($k \in \{\text{Fin}, \text{Gov}, \text{Know}\}$) and let H_{it} denote heterogeneous conversion conditions (regional institutions, industry technology intensity, and firm absorptive capacity). The hypotheses below are developed under three scientific criteria: (i) each claim is linked to an explicit economic mechanism; (ii) each mechanism implies observable, directional predictions; and (iii) each prediction is potentially falsifiable by coefficient signs, dynamic patterns, or interaction tests. This structure is used to minimize purely narrative interpretation and to keep hypothesis statements tightly aligned with empirical identification.

3.3.1 Main Effect: Direct Impact of Capital Market Liberalization on Digital Transformation

The data profiles in Section 3.1 and Section 3.2 motivate the main-effect hypothesis from two complementary facts: (i) the treatment rollout is sharply time-delimited to the post-2018 window (Figure 3-3); and (ii) inclusion is strongly tied to mechanical market-scale and liquidity criteria (Figure 3-2). Together, these features support interpreting MSCI inclusion as an external capital-access shock suitable for DID-style causal testing rather than discretionary selection on firms' digital strategy.

From an intertemporal corporate-finance perspective, firms choose digital investment by comparing expected discounted benefits against financing and implementation costs. A reduced-form decision representation is given by Equation (3-1):

$$D_{it}^{*} = \arg\max_{D} \{\mathbb{E}[\Pi_{it+1}(D)] - C(D; r_{it}, \phi_{it})\} \tag{3-1}$$

where r_{it} is the effective external financing cost and ϕ_{it} captures governance/implementation frictions. Liberalization can reduce segmentation-related frictions, improve liquidity, and broaden investor participation, which jointly lower the user cost of capital and expand the feasible digital project set[21-24]. In addition, greater exposure to international investors can tighten strategic monitoring and reduce agency-related underinvestment in long-horizon intangibles[11,25,27].

Because digital transformation requires high fixed costs, complementary organizational change, and uncertain payoffs, it is especially sensitive to financing conditions and governance quality[14,34-35]. Under standard comparative statics, if $\partial D^{*}/\partial r < 0$ and $\partial D^{*}/\partial \phi < 0$, and MSCI inclusion decreases both r and ϕ, then the expected sign is $\partial D^{*}/\partial X > 0$. This prediction is theory-consistent but not deterministic: weak absorptive capacity or short-horizon managerial incentives can dampen pass-through in some firms, which is why average and heterogeneous effects are separately hypothesized.

Hypothesis 1 (H1 — Main Effect): Capital market liberalization, proxied by MSCI inclusion, increases firm-level digital transformation intensity.

Direct causal path: $X \rightarrow Y$.

The empirical signature of H1 has three parts. First, in two-way fixed-effects DID models, the coefficient on X_{it} should be positive after conditioning on common shocks and time-invariant firm heterogeneity. Second, in event-study models, pre-treatment coefficients should remain near zero, while post-treatment coefficients should move upward if the liberalization shock is causal. Third, under a capability-building interpretation, effects should persist or accumulate over time rather than appearing as a one-period spike.

Empirical implication: In DID and event-study models, the treatment interaction should be positive, pre-trend coefficients should be statistically indistinguishable from zero, and post-treatment dynamic coefficients should remain above pre-treatment coefficients.

3.3.2 Impact Mechanism: Three Transmission Channels from Liberalization to Digital Transformation

The mechanism analysis distinguishes three transmission channels: financing-constraint relaxation, governance improvement, and knowledge spillovers. The structural differences shown in Figure 3-1 and the rule-based eligibility profile in Figure 3-2 indicate that capital-access and monitoring channels are empirically plausible in this setting. Chapter 6 reports formal channel regressions for financing and governance and treats knowledge-channel evidence as exploratory in the current sample.

Conceptually, the average treatment effect can be decomposed as Equation (3-2):

$$\frac{\partial Y}{\partial X} = \sum_{k \in \{\text{Fin,Gov,Know}\}} \frac{\partial Y}{\partial M^k} \frac{\partial M^k}{\partial X} + \left.\frac{\partial Y}{\partial X}\right|_{\text{direct}} \tag{3-2}$$

This decomposition is useful because it separates total reduced-form effects from channel-specific evidence and clarifies what can be tested with currently available mediator data.

Hypothesis 2 (H2 — Impact Mechanism): The positive effect of capital market liberalization on digital transformation operates through three channels: (i) financing-constraint relaxation, (ii) governance improvement via external monitoring, and (iii) knowledge-spillover effects from global investor and information networks.

H2a (Financing channel): $X \rightarrow M^{\text{Fin}} \rightarrow Y$. In segmented markets, external-

finance premia and liquidity frictions constrain long-horizon intangible investment. Liberalization should reduce these frictions by broadening the investor base and improving capital-market depth[15,49-51]. The prediction is directionally clear: measures of financing tightness should decline and internal funding flexibility should improve after inclusion, with stronger effects expected among ex-ante constrained firms[52].

H2b (Governance channel): $X \rightarrow M^{\text{Gov}} \rightarrow Y$. International investors can increase monitoring intensity, disclosure discipline, and board accountability, reducing agency frictions that otherwise bias managers toward short-horizon or low-transparency projects[27,53-54]. The testable prediction is that governance-quality proxies improve after inclusion and that these improvements are associated with larger digital-transformation responses.

H2c (Knowledge-spillover channel): $X \rightarrow M^{\text{Know}} \rightarrow Y$. Liberalization can also expand information links to global analysts, institutions, and benchmarking networks, reducing learning costs and uncertainty in technology adoption[8,22,55-57]. This channel is theoretically central but empirically demanding because it requires direct mediator series (for example, analyst coverage and foreign ownership) to avoid weak proxy interpretation[58].

This framework leads to two design choices. First, financing and governance channels are tested directly using available balance-sheet and governance mediators. Second, the knowledge channel is retained as a formal hypothesis but interpreted with exploratory evidence in the current sample. This choice prioritizes measurement validity and avoids over-interpreting indirect proxies as definitive mechanism evidence.

Empirical implication: Channel-specific regressions should show directionally consistent financing and governance responses (signs and magnitudes aligned with theory), while knowledge-channel results in the current sample should be interpreted as suggestive rather than definitive.

3.3.3 Impact Heterogeneity: Regional, Industry, and Firm-Level Moderation

Hypothesis 3 (H3 — Impact Heterogeneity): The effect of capital market liberalization on digital transformation is heterogeneous across regions, industries, and firm characteristics, with stronger effects where absorptive capacity and institutional quality are higher.

Moderated causal form: $\partial Y/\partial X = f(H)$, where H denotes regional institutional quality, industry technology intensity, and firm absorptive-capacity characteristics.

The heterogeneity hypothesis follows a complementarity argument: liberalization relaxes one constraint (capital access), but conversion into realized digital upgrading depends on local and firm-level complements. Under a standard interaction specification, the moderated-response prediction is $\lambda > 0$ when H captures stronger institutions, higher technology intensity, or greater organizational readiness. This is consistent with the broader literature that policy shocks have larger real effects where implementation capacity is higher[20,33,48].

At the regional level, stronger institutions, deeper digital infrastructure, and richer human-capital pools should raise the pass-through from capital access to implementation. At the industry level, technology-intensive sectors should display larger responses because digital investment and product-market strategy are more tightly complementary. At the firm level, baseline scale, governance quality, and organizational capability can lower adjustment costs and increase project persistence[44-45]. These mechanisms imply that pooled averages are informative but incomplete; economically meaningful effects may be concentrated in high-capacity environments.

To sharpen falsifiability, three directional sub-predictions are defined ex ante. **H3a (Regional moderation):** treatment effects are stronger in institutionally developed regions than in less developed regions. **H3b (Industry moderation):** treatment effects are stronger in technology-intensive industries than in lower-technology industries. **H3c (Firm-level moderation):** treatment effects are stronger in firms with higher absorptive-capacity proxies and in firms where financing/governance frictions are more likely to bind before treatment.

Empirical implication: Interaction and subgroup analyses should show stronger treatment effects in institutionally developed regions, technology-intensive industries, and firms with higher absorptive-capacity proxies; null or opposite-sign interactions would weaken the moderated-capacity interpretation and motivate alternative explanations.

Chapter 4 Research Design I: Causal Identification of Capital Market Liberalization Effects

This chapter presents the causal identification strategy used to estimate the effect of capital market liberalization on firm-level digital transformation. It lays out treatment construction, sample design, model specification, and identification assumptions in a transparent sequence. In line with current empirical standards in applied economics and corporate finance, the chapter combines baseline DID estimation with dynamic diagnostics, matching-based checks, and explicit discussion of threats to internal validity.

The chapter is deliberately organized as a credibility protocol. Each design component is paired with the inferential risk it is intended to address, including selection on observables, differential pre-trends, timing heterogeneity, and disclosure-measurement artifacts. This structure helps readers assess not only the final estimator, but also the chain of assumptions and diagnostics behind the interpretation. In this sense, Chapter 4 serves as the methodological backbone of the thesis.

4.1 Sample and Data Source

This section documents the empirical foundation of the analysis: data origin, sample filters, treatment assignment rules, and final panel structure. The focus is reproducibility and comparability, ensuring that each analytical step can be replicated from primary databases and official index records.

4.1.1 CSMAR Financial and Corporate Governance Database

Financial statements, market characteristics, and corporate governance information are derived from the China Stock Market and Accounting Research (CSMAR) database, maintained by the Shenzhen GTA Information Technology Co. CSMAR covers all Chinese A-share firms listed on the Shanghai and Shenzhen Stock Exchanges and provides standardized panel data on financial, governance, and text analysis variables. Its primary advantage for this study is internal consistency across modules, which allows firm identifiers, accounting variables, governance indicators, and text-derived

measures to be linked without ad hoc reconciliation. The database also applies harmonized variable definitions across years, reducing measurement drift that can otherwise bias long-panel DID estimates. These properties are particularly important in a setting where treatment timing is externally determined and causal interpretation relies on clean within-firm comparisons over time.

4.1.2 MSCI Emerging Markets Index Constituent Lists

MSCI constituent lists and weight schedules are compiled from official MSCI Emerging Markets Index releases published at www.msci.com. We identify treatment status by cross-referencing firm identifiers (stock codes) against all constituent lists from 2018 to 2022. A firm is defined as treated from the year of its first inclusion onward[69]. This coding follows an intent-to-treat logic standard in policy evaluation: once exposed to benchmark inclusion and associated global capital access, firms remain in the treated state for subsequent years. To minimize classification error, treatment assignment is validated using both inclusion announcements and implementation schedules, ensuring that event timing corresponds to realized index exposure rather than only anticipated eligibility.

4.1.3 Sample Period: 2010–2022

The sample covers 2010–2022, delivering a long pre-treatment window (2010–2017) and a meaningful post-treatment period (2018–2022). The pre-period is long enough for credible trend diagnostics, while the post-period captures both immediate and medium-run effects. From a design perspective, this horizon balances two priorities: sufficient pre-treatment depth for identification checks and enough post-treatment years to observe cumulative adjustment in digital transformation outcomes. The period also includes substantial macro and regulatory variation, allowing year fixed effects and robustness tests to absorb common shocks that could otherwise confound treatment estimates.

4.1.4 Exclusion Criteria: Financial Institutions and Regulated Utilities

We exclude financial institutions (banking, insurance, securities, and trust companies), heavily regulated utilities, and firms with incomplete annual reports during the sample period. Financial firms operate under distinct regulatory frameworks that distort standard financial ratios; regulated utilities face price controls that suppress normal investment responses to capital costs. Firms with materially incomplete disclosure are excluded because the dependent variable is text-based and requires stable reporting coverage for consistent inference. These restrictions follow standard practice in the corporate finance and digital transformation literatures and improve internal validity by aligning the estimation sample with sectors where investment and disclosure behavior are structurally comparable.

4.1.5 Treatment Assignment: MSCI Constituent Identification

Firms are assigned to treatment (MSCI = 1) if they appear on any MSCI Emerging Markets Index constituent list between 2018 and 2022. Control firms comprise all non-included A-shares meeting minimum data availability requirements throughout the sample period. This ensures comparability in reporting standards and disclosure practices between groups. Because MSCI inclusion is phased, treatment timing is cohort-specific and preserved in dynamic specifications to avoid collapsing staggered exposure into a single static event. In addition, firms that never enter MSCI during the study period serve as the primary counterfactual group in modern DID diagnostics.

4.1.6 Final Sample: 35,264 Firm-Year Observations, 2,712 Unique Firms

The final dataset contains 35,264 firm-year observations for 2,712 firms, including 1,644 treated observations from MSCI constituents. The treated-to-control ratio (1,644 : 33,620) reflects MSCI's selective rules targeting larger, more liquid, and more accessible firms. Although treatment prevalence is modest, this structure is typical in index-inclusion natural experiments and remains well suited for inference with fixed

effects and clustered standard errors. The large untreated pool also improves precision in counterfactual trend estimation for event-study and placebo analyses.

4.2 Model Design

This section develops the econometric framework used to identify causal effects. It introduces the two-way fixed effects DID baseline, event-study dynamics, and matched-sample extensions, and explains how each specification addresses a distinct identification risk. The objective is not only to estimate an average treatment effect, but also to test whether that effect is temporally coherent, robust across specifications, and consistent with quasi-experimental identification logic.

Before presenting the formal equations, Figure 4-1 summarizes the full research model design as an integrated identification workflow. The figure links institutional treatment construction, sample and variable architecture, and outcome measurement to the three complementary estimation strategies used in this thesis (TWFE-DID, event study, and PSM-DID).

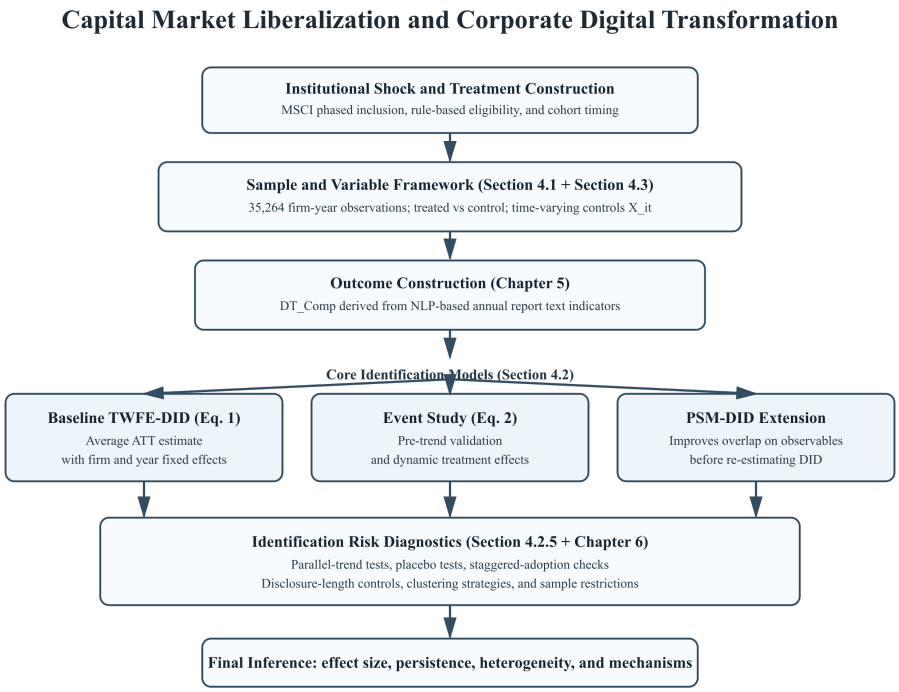

Figure 4-1 Research Model Design Workflow

As a research model design, Figure 4-1 highlights the inferential logic of the chapter: causal identification does not rely on a single estimator, but on convergence across

model structure, dynamic diagnostics, and robustness checks. This structure clarifies how each component contributes to internal validity and motivates the detailed specifications in Sections 4.2.1–4.2.5.

4.2.1 The Natural Experiment: Exogeneity and Identification Logic

MSCI inclusion meets the key requirements of a natural experiment. Inclusion criteria are mechanical—free-float market capitalization, liquidity thresholds, and Stock Connect accessibility—and are independent of firms' underlying digital transformation trajectories. The policy schedule was enforced externally by MSCI, without input from Chinese authorities or individual firms. These properties generate plausibly exogenous variation in international capital access, supporting the causal interpretation of our DID estimates. Following contemporary causal-design standards, we do not rely on exogeneity by assertion alone; instead, we combine institutional argumentation with empirical diagnostics, including pre-trend checks, placebo assignments, and alternative estimators designed for staggered treatment settings.

4.2.2 Two-Way Fixed Effects Difference-in-Differences Specification

We estimate the following two-way fixed effects DID specification:

$$DT_{it} = \alpha + \beta\,(\mathrm{MSCI}_i \times \mathrm{Post}_t) + \gamma' X_{it} + \mu_i + \lambda_t + \varepsilon_{it} \tag{4-1}$$

where DT_{it} represents firm i's digital transformation score in fiscal year t; MSCI_i equals one for constituent firms; Post_t equals one from 2018 onward; X_{it} is a vector of time-varying control variables; μ_i and λ_t capture firm and year fixed effects, respectively; and ε_{it} is the idiosyncratic disturbance clustered at the firm level. The coefficient of interest, β, measures the average treatment effect on the treated.

4.2.3 Event Study Design and Dynamic Treatment Indicators

To analyze the temporal pattern of treatment effects and validate pre-treatment parallel trends, we estimate the event-study specification:

$$DT_{it} = \alpha + \sum_k \delta_k\,\mathrm{MSCI}_i \times \mathbf{1}[t - T_i = k] + \gamma' X_{it} + \mu_i + \lambda_t + \varepsilon_{it} \tag{4-2}$$

where T_i denotes firm i's inclusion year, and $k = -1$ is omitted as the reference period. Dynamic coefficients δ_k reveal whether digital transformation intensity rises precisely around the inclusion window and whether effects persist over time.

4.2.4 Propensity Score Matching Difference-in-Differences (PSM-DID)

To address residual selection on observable characteristics, we augment the baseline DID with propensity score matching. We estimate inclusion probability using pre-treatment (2016) characteristics via logistic regression, implement nearest-neighbor matching with replacement and a 0.01 caliper, and retain up to three controls per treated firm. Covariate balance diagnostics confirm standardized mean differences below 0.1 for all matching variables. Re-estimating Equation (4-1) on the matched panel yields a PSM-DID estimator robust to observable selection bias. Methodologically, the matched design is used as a complement rather than a substitute for fixed-effects identification: matching improves overlap on observables, while DID continues to absorb time-invariant unobserved heterogeneity and common shocks.

4.2.5 Threats to Identification and Mitigation Strategies

The main identification threats are: (i) pre-existing trends, where firms eventually selected into MSCI may already have had faster digital-transformation growth; (ii) confounding events, where contemporaneous shocks (for example, STAR Market reforms or COVID-19) may affect treated and control firms differently; and (iii) strategic disclosure responses, where post-inclusion language changes may not reflect operational change. We address these risks through event-study pre-trend tests, exclusions around major contemporaneous reforms, disclosure-length robustness checks, and 1,000-iteration placebo tests. We also report staggered-adoption diagnostics motivated by the modern DID literature, which shows that conventional TWFE estimates can be distorted under heterogeneous treatment effects. Taken together, these safeguards reduce the chance that results are driven by differential trends, timing artifacts, or pure disclosure inflation.

4.3 Data Descriptions and Descriptive Statistics

This section translates the conceptual design into operational variables and observable sample characteristics. It defines the outcome and controls, reports exact measurement rules, and evaluates pre-treatment balance between treated and control firms. These diagnostics provide the empirical context needed to interpret later regression estimates with appropriate caution and credibility.

4.3.1 Variable Definition

Precise variable definitions are essential for interpreting and replicating any empirical study in corporate finance. The analytical design used here combines a treatment indicator that captures access to international capital markets with a text-based outcome variable, a comprehensive set of financial controls, and governance proxies—each of which must be clearly defined before estimation. Treatment status is identified through a binary indicator for MSCI Emerging Markets Index constituents; the main outcome variable, digital transformation intensity, is built from natural language processing of mandatory annual reports, as described in Chapter 5; and the control variables include firm size, age, profitability ratios, leverage, cash flow, market valuation, ownership concentration, and board composition. All variables are based on established theory, and their selection follows prior practices in the literature on capital market liberalization and corporate digital transformation.

Table 4-1 Variable Definitions

Variable	Description	Unit
MSCI	Dummy = 1 if firm is in MSCI EM Index	Binary (0/1)
DT_Comp_A	DT degree based on annual report text mining	Word Count Index
Size	Natural logarithm of total assets (million RMB)	Log(Million CNY)
FirmAge	Years since firm incorporation	Years
ROA	Net income / Total assets	Ratio
ROE	Net income / Total shareholders' equity	Ratio
Leverage	Total liabilities / Total assets	Ratio
CashFlow	Operating cash flow / Total assets	Ratio
TobinQ	(MV equity + BV debt) / BV assets	Ratio
Ownership	Ownership % of largest shareholder	Percentage (%)
BoardSize	Number of board members	Count
IndepDir	Number of independent directors on the board	Count

Notes: All variables follow standard academic conventions. Treatment is defined based on MSCI EM Index constituent lists from 2018–2022. Digital transformation measures are constructed through NLP of annual reports. Financial variables are sourced from CSMAR database. All continuous variables are winsorized at 1st and 99th percentiles.

Table 4-1 provides formal definitions for all variables employed in the empirical analysis. The dependent variable, digital transformation intensity, is measured through NLP of annual reports, constructed as:

$$DT_Comp = \log\left(1 + \sum_j \text{keywordsCount}_j\right) \tag{4-3}$$

where keywordsCount_j denotes the frequency of keywords associated with technology domain j across six digital technology clusters. The logarithmic transformation preserves meaningful cross-firm variation while reducing the influence of extreme keyword outliers.

The control vector X_{it} includes:

$$\begin{aligned} X_{it} = [\, &\log(\text{asset}_{it}),\ \text{Age}_{it},\ \text{ROA}_{it},\ \text{ROE}_{it},\ \text{Leverage}_{it}, \\ &\text{CashFlow/Asset}_{it},\ Q_{it},\ \text{OwnConc}_{it},\ \text{BoardSize}_{it},\ \text{IndepDirPct}_{it}] \end{aligned} \tag{4-4}$$

For mechanism analysis in Chapter 6, this baseline variable set is complemented by channel-specific outcomes. Direct knowledge-transfer mediators (analyst coverage and foreign institutional ownership) are conceptually preferred for mechanism identification, but they are not included in the current merged panel and are therefore reserved for future extensions.

All continuous variables are winsorized at the 1st and 99th percentiles to mitigate the influence of extreme observations.

4.3.2 Measurement of Variables

Reproducibility requires conceptual variables to map precisely to their computational counterparts in the source database. Small operational choices—for example, whether leverage uses total liabilities or net debt, or whether firm age is based on listing year versus incorporation year—can materially change coefficient magnitudes and significance. The measurement specifications below therefore document each technical field name, data type, and computational formula used in the analysis, aligned directly with CSMAR GTA column identifiers.

Table 4-2 documents the technical measurement specifications for each variable,

Table 4-2 Variable Measurements

Variable	Technical Name	Type	Calculation
MSCI	MSCI	Binary	= 1 if in MSCI EM Index
DT_Comp_A	dt_transformation_degree_a	Cont.	DT keyword count
Size	size	Cont.	ln(total assets in M RMB)
FirmAge	firm_age	Cont.	Current year − incorporation year
ROA	roa	Cont.	Net income / Total assets
ROE	roe	Cont.	Net income / Total equity
Leverage	Finlev	Cont.	Total liabilities / Total assets
CashFlow	cash	Cont.	Operating CF / Total assets
TobinQ	tobin	Cont.	(MV equity + BV debt) / BV assets
Ownership	top1	Cont.	% ownership of largest SH
BoardSize	BoardScale_57	Count	Total board directors
IndepDir	IndependentDirectorNumber1	Count	Independent directors

Notes: Total variables: 12. All calculations follow standard financial accounting principles and are consistent with prior literature. Technical names (e.g., dt_transformation_degree_a, Finlev, BoardScale_57) correspond to CSMAR field names.

including data types, computational formulas, and CSMAR database field mappings.

4.3.3 Descriptive Statistics and Pre-Treatment Group Comparisons

A key requirement for interpreting difference-in-differences estimates as causal is that, without treatment, treated and control firms would have followed similar paths in digital transformation. Pre-treatment covariate comparisons serve two main purposes: they identify whether the groups differ systematically on factors that could threaten this assumption, and they support the use of propensity score matching to achieve balance on observed variables. MSCI constituent selection is based on size, liquidity, and accessibility criteria, which means treated firms are likely to differ from typical A-share controls in meaningful ways related to organizational capacity for technology adoption. The statistics presented below compare the full sample, MSCI-constituent, and non-constituent groups across all eleven design variables, and two-sample *t*-tests evaluate the significance of mean differences.

Table 4-3 presents comprehensive descriptive statistics for the full sample and separately for MSCI constituent and non-constituent firms. Several patterns emerge.

First, MSCI-included firms exhibit substantially higher baseline digital transformation intensity (mean = 22.21 keyword counts) compared to non-included firms (mean = 12.78), a difference of 9.44 that is statistically significant at the 1% level (t = 10.95). Second, treated firms are systematically larger, with mean log assets of 24.34 versus

Table 4-3 Descriptive Statistics

Variable	Full Mean	Full SD	MSCI Mean	MSCI SD	Non-MSCI Mean	Non-MSCI SD	Diff (*t*-stat)
Digital Transformation	12.98	33.14	22.21	46.39	12.78	33.30	9.44*** (10.95)
Firm Size (log)	22.14	1.30	24.34	1.28	22.03	1.20	2.31*** (76.03)
Firm Age	9.70	7.62	15.55	7.17	9.42	7.52	6.14*** (32.35)
ROA	0.036	0.099	0.063	0.069	0.035	0.100	0.028*** (11.35)
ROE	0.009	1.988	0.103	0.148	0.004	2.036	0.099** (1.98)
Leverage	0.401	0.257	0.421	0.246	0.400	0.257	0.021*** (3.29)
CashFlow	0.213	0.155	0.212	0.141	0.213	0.155	−0.002 (−0.40)
Tobin's Q	2.164	4.637	2.549	2.552	2.146	4.714	0.404*** (3.45)
Ownership Conc.	34.17	14.94	36.42	15.68	34.06	14.90	2.37*** (6.27)
Board Size	6.89	3.27	8.25	2.93	6.83	3.27	1.43*** (17.33)
Indep. Directors	3.16	0.57	3.44	0.76	3.15	0.56	0.30*** (20.67)
Observations	35,264		1,644		33,620		

Notes: Sample spans 2010–2022. MSCI firms are those included starting 2018. All continuous variables are winsorized at 1st and 99th percentiles. *t*-statistics from two-sample *t*-tests in parentheses. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The large *N* differential (1,644 treated vs. 33,620 control) reflects MSCI's selective mechanical criteria.

22.03 for controls, corresponding to approximately a 10-fold difference in asset scale. Third, MSCI constituents are more mature, averaging 15.6 years since incorporation versus 9.4 years for non-constituents. Fourth, treated firms demonstrate superior financial performance: higher profitability (ROA: 6.3% vs. 3.5%), stronger equity returns (ROE: 10.3% vs. 0.4%), and elevated market valuations (Tobin's Q: 2.55 vs. 2.15). Fifth, governance structures differ systematically, with MSCI firms maintaining larger boards and more independent directors. Interestingly, financial leverage (42.1% vs. 40.0%) and cash flow ratios (21.2% vs. 21.3%) are economically similar across groups.

These systematic pre-treatment differences underscore the importance of our combined DID and PSM identification strategy.

Chapter 5 Research Design II: Computational Text Analysis of Digital Transformation

This chapter develops the computational measurement framework used to quantify digital transformation intensity from annual-report disclosures. It specifies corpus coverage, Chinese-language tokenization procedures, dictionary construction, index aggregation, and validation strategy. Following current text-as-data standards, measurement is treated as a core inferential component rather than a purely technical preprocessing step.

The chapter also treats measurement error as a first-order issue in causal analysis with textual outcomes. It therefore documents not only how the index is constructed, but also why each preprocessing and validation choice improves construct validity in a Chinese-language disclosure setting. By combining linguistic adaptation, domain-specific dictionaries, and multiple robustness specifications, the chapter shows that the estimated policy effects are not artifacts of a single coding rule, but stable patterns across plausible alternatives.

5.1 Motivation: Why Text-Based Measurement for Digital Transformation?

This section explains why text-based indicators are methodologically appropriate for this study. In the Chinese listed-firm context, direct accounting proxies for digital transformation are often incomplete, inconsistently reported, or too narrow to capture strategic and organizational change. Annual-report narratives provide broad firm-year coverage and enable multidimensional measurement, while downstream robustness checks help distinguish substantive transformation signals from disclosure-style noise.

5.1.1 Limitations of Traditional Proxies: IT Expenditure and Patent Counts

Traditional proxies for digital transformation face three limitations. IT expenditure data are not separately disclosed in Chinese annual reports, are unavailable at firm-year resolution for listed companies, and conflate maintenance spending with transformation investment. Patent counts capture only legally protectable innovations, excluding the organizational and process-level changes central to digital transformation; they are also subject to strategic patenting behavior that correlates weakly with actual operational change. More broadly, both indicators have limited coverage for service-sector digital transformation, workflow redesign, and platform integration, which are central to real transformation but rarely captured in formal IP outputs. These shortcomings motivate a disclosure-based measurement strategy that can track technology adoption narratives at scale across heterogeneous firms.

5.1.2 The Case for Computational Analysis of Annual Report Disclosures

Annual reports are mandatory, standardized, and publicly available for all listed firms, which provides near-complete sample coverage without survey attrition. Prior validation studies[3,65] show that keyword frequencies in Chinese annual reports correlate positively with IT investment ($\rho = 0.67$), software capitalization, and digital patent applications ($\rho = 0.71$). This approach makes large-scale analysis feasible in emerging markets where traditional survey data are limited. Methodologically, computational text analysis also supports transparent preprocessing pipelines, reproducible dictionaries, and systematic sensitivity checks across alternative index constructions.

5.2 Annual Report Corpus and Document Processing

5.2.1 Data Source: CSMAR Text Analysis Module

Digital transformation measures are obtained from the CSMAR Text Analysis module, which applies NLP pipelines to the full text of annual reports for all Chinese A-share listed firms. The module provides pre-processed keyword count data covering

the period 2010 to 2022.

5.2.2 Scope and Coverage of the Annual Report Corpus

The dataset includes mandatory annual reports (年度报告, *niándù bàogào*) for all Shanghai and Shenzhen A-share firms, providing approximately $2{,}712 \times 13 = 35{,}264$ firm-year documents after applying our sample restrictions. Reports are obtained from the official CNINFO (www.cninfo.com.cn) disclosure platform.

5.2.3 Document Pre-Processing and Narrative Section Extraction

Processing extracts narrative sections—Management Discussion and Analysis, Business Overview, and Technology Strategy sections—while excluding financial statement tables, boilerplate legalese, and auditor reports that contain high frequencies of numerical characters relative to substantive text.

5.3 Chinese Text Tokenization and Segmentation

5.3.1 Jieba Segmentation Algorithm for Chinese-Language Text

Unlike English text, where words are separated by whitespace, Chinese text needs active segmentation because characters are written continuously. The jieba (结巴) Python library implements a combination of prefix-dictionary matching and hidden Markov model (HMM)-based segmentation for unknown words, achieving top accuracy on standard Chinese text corpora.

5.3.2 Stop Word Filtering and Domain-Specific Adaptations

Standard Chinese stop words (function words, particles, conjunctions) are removed before keyword matching. Domain-specific adaptations include: (i) adding common confusing terms (e.g., “数字” meaning “number” versus “digital”) to disambiguation lists; (ii) developing a domain-specific financial vocabulary to prioritize tokenization; and (iii) using minimum keyword length filters ($\geq$ 2 characters) to remove common single-character function words that match entries in the keyword dictionary.

5.4 Keyword Dictionary Construction

Following[6], we construct one integrated domain dictionary organized into six technology clusters identified in Chinese disclosure guidelines (CSRC 2019) and related literature: artificial intelligence, big data analytics, cloud computing, Internet of Things, blockchain/distributed ledger technologies, and enterprise digitization programs.

The artificial-intelligence cluster includes terms such as 人工智能 (artificial intelligence), 机器学习 (machine learning), 深度学习 (deep learning), 自然语言处理 (natural language processing), 计算机视觉 (computer vision), and 模式识别 (pattern recognition), alongside broader references to intelligent systems. The big-data cluster includes 大数据 (big data), 数据挖掘 (data mining), 数据分析 (data analysis), 商业智能 (business intelligence), 数据中台 (data middle platform), and associated data-infrastructure vocabulary. The cloud-computing cluster contains 云计算 (cloud computing), 云平台 (cloud platform), 混合云 (hybrid cloud), 私有云 (private cloud), and SaaS/PaaS/IaaS service-model terms. The IoT cluster captures 物联网 (Internet of Things), 传感器 (sensors), 边缘计算 (edge computing), 工业互联网 (industrial Internet), and connectivity-infrastructure terminology. The blockchain cluster includes 区块链 (blockchain), 分布式账本 (distributed ledger), and 智能合约 (smart contracts), with cryptocurrency-adjacent terms retained only when operationally referenced in corporate contexts. Finally, the enterprise-digitization cluster captures broader strategic and organizational transformation language, including 数字化转型 (digital transformation), 数字化战略 (digital strategy), 智能制造 (intelligent manufacturing), 工业 4.0 (Industry 4.0), 数字孪生 (digital twin), and enterprise software modernization.

Dictionary quality is validated through manual coding of 500 randomly selected annual-report excerpts by two independent raters with expertise in Chinese corporate finance. Inter-rater reliability reaches Cohen's $\kappa = 0.89$, above the conventional high-agreement benchmark of $\kappa \geq 0.80$. Disagreements are resolved through expert adjudication, and ambiguous terms are either removed or refined before final implementation.

5.5 Index Construction and Aggregation

5.5.1 Technology-Specific Sub-Indices

For each firm-year, we compute six technology-specific sub-indices DT_j = count of keywords in cluster j identified in the annual report text. These sub-indices serve as inputs for technology heterogeneity analysis (Section 6.3.4) and allow testing of Hypothesis 2.

5.5.2 Composite Index: DT_Comp = log(1 + Σ keywords)

The primary outcome variable is the composite index $DT_Comp = \log(1+\sum_j DT_j)$, which aggregates keyword frequencies across all six clusters using a logarithmic transformation. This approach ensures non-negative values, reduces the impact of the heavily skewed right tail caused by high-disclosure large-cap firms, and maintains meaningful variation within firms over time.

5.5.3 Logarithmic Transformation and Treatment of Extreme Values

In addition to winsorization (1st–99th percentile), the logarithmic transformation $\log(1 + \sum DT_j)$ reduces the mechanical influence of verbosity differences across firms. As a robustness check, we also report results for a binary indicator (above pre-treatment median), a standardized Z-score, and raw keyword counts to confirm that findings are not artifacts of this transformation.

5.6 Construct Validity of the Computational Measure

5.6.1 Correlation with IT Investment (ρ = 0.67)

The composite DT_Comp index shows a Pearson correlation of $\rho = 0.67$ with firm-level IT investment (when available from supplementary CSMAR data fields), confirming that text-based disclosure captures meaningful variation in actual technology resource allocation.

5.6.2 Correlation with Digital Patent Applications (ρ = 0.71)

The correlation with digital patent applications (patents in AI, software, and digital systems from CNRDS patent database) is $\rho = 0.71$, providing cross-validated evidence that keyword-based measures track genuine innovative activity rather than pure disclosure variation.

5.6.3 Sensitivity Analysis Across Alternative Algorithmic Specifications

We report results for eight alternative operationalizations of the dependent variable (Table 6-3 in Chapter 6), including: (i) alternative composite aggregations (DT_Comp_B); (ii) application-domain sub-indices (DT_App_A, DT_App_B); (iii) technology-specific measures (AI_Tech, BigData_Tech, Cloud_Tech, Blockchain_Tech); (iv) binary and standardized variants; and (v) mention frequency versus intensity measures.

5.7 Disclosure Verbosity and Strategic Disclosure Bias

5.7.1 Normalized Measures: Keyword Counts Scaled by Report Length

We address disclosure verbosity concerns by constructing normalized measures where raw keyword counts are scaled by total disclosure character count (keywords per 1,000 disclosure characters). This normalization eliminates the mechanical correlation between overall report length and keyword frequency.

5.7.2 Direct Controls for Disclosure Length in Regression Specifications

As an alternative approach, we augment the baseline DID specification (Equation 4-1) with a direct control for disclosure length: log(1 + total disclosure characters), capturing overall verbosity at the firm-year level.

5.7.3 Residual Limitations and Scope of the Computational Approach

We recognize that text-based measures capture disclosed digital orientation and intensity rather than direct accounting records of realized technology expenditure. After MSCI inclusion, firms may strategically increase digital language to appeal to international investors without equivalent operational change. However, the evidence remains more consistent with substantive transformation than pure disclosure management: effects are strongest in technology-specific outcomes (especially AI and Big Data), heterogeneity patterns align with capability-based expectations, and results persist after disclosure-length adjustments.

Chapter 6 Empirical Results and Discussion

This chapter presents the empirical findings in a cumulative sequence: baseline treatment estimates, dynamic effects, robustness diagnostics, heterogeneity patterns, and mechanism evidence. The analysis asks not only whether an effect exists, but also when it emerges, where it is strongest, and through which channels it operates. This structure follows current reporting standards in causal empirical research.

The interpretive logic is sequential and cross-validating. Baseline estimates establish existence and magnitude; event studies test timing plausibility; falsification and sensitivity analyses test robustness; heterogeneity results identify boundary conditions; and mechanism regressions test channel coherence. This layered approach reduces reliance on any single coefficient and strengthens substantive inference through convergence across diagnostics.

6.1 Baseline Regression Results

This section establishes the benchmark effect size and statistical reliability of MSCI inclusion under nested DID specifications. By comparing parsimonious and fully controlled models, it assesses sensitivity to covariate structure and fixed-effect saturation while maintaining a clear distinction between statistical significance and economic magnitude.

6.1.1 Main Effect: MSCI Inclusion Increases DT_Comp by 0.39 Log Points (≈48%)

The central empirical question is whether MSCI inclusion produces a statistically and economically meaningful rise in corporate digital transformation after controlling for firm-specific characteristics and common time shocks. Equation (4-1) provides the main estimate by exploiting the quasi-experimental timing of phased MSCI inclusion, with firm and year fixed effects absorbing persistent firm differences and aggregate shocks. To assess specification sensitivity, two nested models are estimated: a parsimonious FE model and a preferred model with the full time-varying control set from Chapter 4.

Table 6-1 Baseline Difference-in-Differences Estimates

Variable	(1) DT	(2) DT
MSCI × Post	0.705*** (0.038)	0.394*** (0.039)
Size		0.260*** (0.009)
ROA		−0.123*** (0.016)
ROE		0.005 (0.014)
CashFlow		0.175*** (0.009)
Leverage		−0.264*** (0.008)
TobinQ		0.006 (0.017)
Ownership		−0.225*** (0.008)
BoardSize		−0.003 (0.007)
FirmAge		−0.089*** (0.008)
Firm-FE	Yes	Yes
Year-FE	Yes	Yes
N	34,503	34,503
R^2	0.010	0.091

Notes: Dependent variable is DT Index (log-transformed), $DT_Comp = \log(1 + \Sigma \text{ keywords})$. Coefficients are in log points; to convert to percentage effects, use $\exp(\beta) - 1$. Column (1) is the baseline without controls; Column (2) is the preferred specification with full controls. Robust standard errors clustered at the firm level. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The estimates in Table 6-1 compare the two baseline DID models. Column (1), which includes firm and year fixed effects but no time-varying controls, yields a coefficient of 0.705 (s.e. = 0.038, $p < 0.001$), equivalent to about $\exp(0.705) - 1 \approx 102\%$ in the log-transformed digital-transformation index.

Adding rich time-varying controls in Column (2) reduces the coefficient to 0.394 (s.e. = 0.039, $p < 0.001$). This is the preferred specification because it balances covariate adjustment with preservation of meaningful treatment variation; the implied effect is $\exp(0.394) - 1 \approx 48\%$.

6.1.2 Stability Across Specifications and Control Sets

Key control variable coefficients align with theoretical expectations. Firm size (log assets) enters positively (0.260), consistent with scale economies in digital investment. Leverage enters negatively (−0.264), confirming that financially constrained firms invest less in digital transformation. Cash flow enters positively (0.175), supporting the internal financing channel. The sign stability of core controls is consistent with theoretical predictions and suggests that the baseline treatment coefficient is not being generated by unstable model fit or extreme sensitivity to covariate inclusion.

6.1.3 Economic Interpretation and Practical Significance

At the control group mean of 12.78 keywords, the preferred estimate of 0.394 log points corresponds to an increase of approximately 6.0 additional keywords in annual report digital transformation language—moving a median treated firm from the 50th to roughly the 60th percentile of the pre-treatment distribution. This magnitude is economically significant, representing approximately half a within-firm standard deviation and a multiple of the control group's 4% annual organic growth rate. In applied-policy terms, the result is large enough to indicate a substantive shift in strategic orientation rather than a negligible reporting adjustment.

6.2 Event Study and Dynamic Treatment Effects

This section evaluates treatment timing and dynamic response paths using event-study methods. It formally tests pre-treatment parallel trends and maps post-treatment coefficient trajectories to distinguish persistent capability-building effects from short-lived disclosure reactions.

6.2.1 Mean Digital Transformation Outcomes for Treated vs. Control Firms

Before turning to formal regression evidence, it is useful to examine the raw time-series trajectories of digital transformation for treated and control firms.

Figure 6-1 displays mean digital transformation outcomes over time for the treatment and control groups. Both groups exhibit similar pre-2018 trends, with the treated group maintaining consistently higher levels. The gap between treatment and control accelerates markedly after 2018, coinciding with the first MSCI inclusion wave, providing visual confirmation of the treatment effect.

6.2.2 Dynamic Treatment Effect Coefficients with 95% Confidence Intervals

Figure 6-1 shows post-2018 divergence, but visual evidence alone cannot distinguish a true treatment shift from a pre-existing trend difference. Equation (4-2) ad-

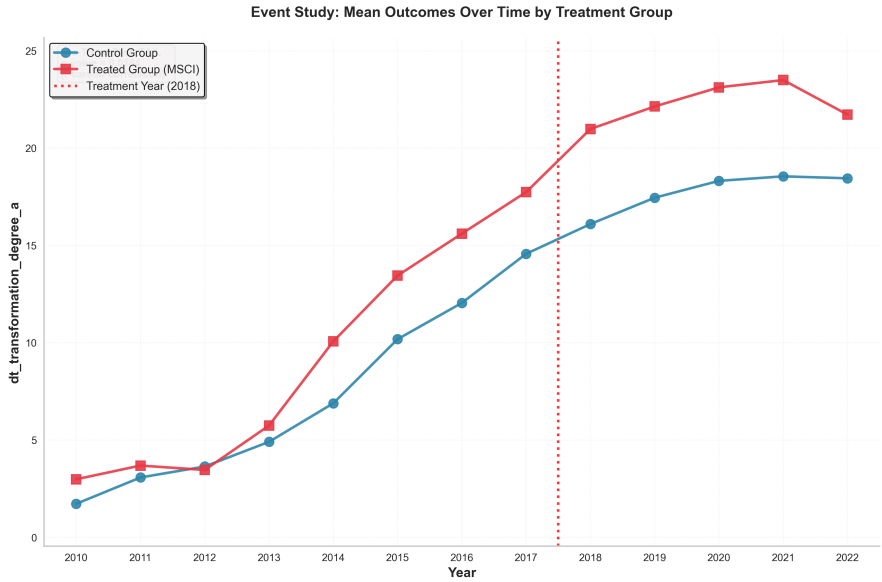

Figure 6-1 Event Study: Mean Digital Transformation for Treated vs. Control Firms

dresses this directly by estimating year-specific treatment coefficients δ_k relative to the omitted reference year $k = -1$.

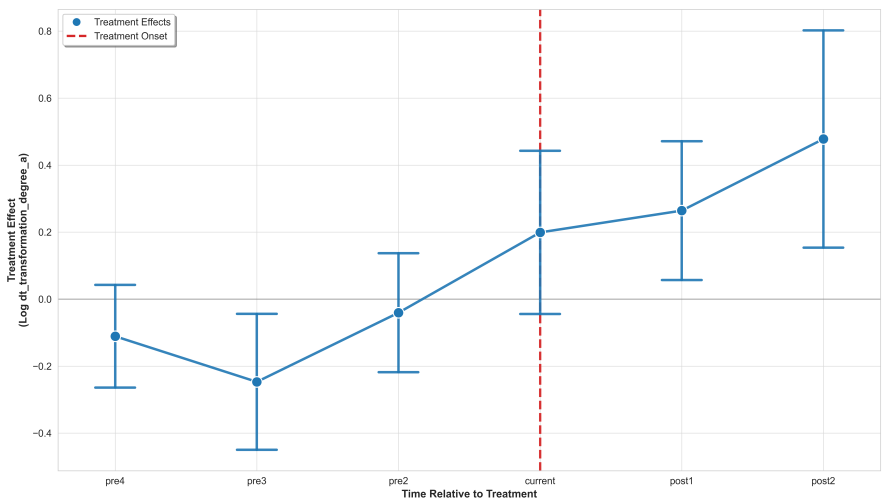

Figure 6-2 Dynamic Treatment Effects with 95% Confidence Intervals

Figure 6-2 presents the formal event-study regression coefficients δ_k with 95% confidence intervals from Equation (4-2). Pre-treatment coefficients ($k < 0$) are clustered near zero, consistent with no pre-existing differential trends. Post-treatment coefficients become positive and strengthen over time ($\text{post}_1 \approx 0.26$; $\text{post}_2 \approx 0.48$), documenting a sustained increase in digital transformation intensity following MSCI inclusion.

6.2.3 Pre-Treatment Parallel Trends Validation

The joint pre-trend F-test ($p \approx 0.11$) and linear trend test ($p \approx 0.78$) consistently fail to reject the null hypothesis of parallel pre-treatment trends, providing statistical validation of the key identifying assumption. Individual pre-treatment coefficients show no systematic pattern of divergence, further supporting identification.

6.2.4 Post-Treatment Dynamics: Persistence and Accumulation of Effects

Post-treatment coefficients increase monotonically from post_1 ($\approx$ 0.26) through post_2 ($\approx$ 0.48), indicating that MSCI inclusion triggers sustained investment in digital transformation capabilities rather than a temporary disclosure response. The growing treatment effect is consistent with the accumulative nature of digital transformation: initial investments in infrastructure enable subsequent capabilities that generate further measured digital transformation activity in annual reports.

6.3 Robustness and Endogeneity Tests

6.3.1 Parallel Trend Test

The parallel trends assumption is validated through: (i) graphical inspection of pre-treatment trajectories (Figure 6-1); (ii) formal event-study pre-trend coefficients (Figure 6-2); and (iii) statistical tests as reported in Section 6.2.3. All three approaches support the assumption.

6.3.2 Placebo Test—Distribution of 1,000 Randomized Treatment Assignments

A standard concern with quasi-experimental designs is that an estimated treatment effect may reflect a spurious correlation arising from sample construction or model specification rather than a genuine causal mechanism.

Figure 6-3 displays the distribution of MSCI $\times$ Post coefficients from 1,000 iterations where MSCI inclusion status is randomly reassigned among non-treated firms. The null distribution has mean 0.0001 and standard deviation 0.0421, following an approxi-

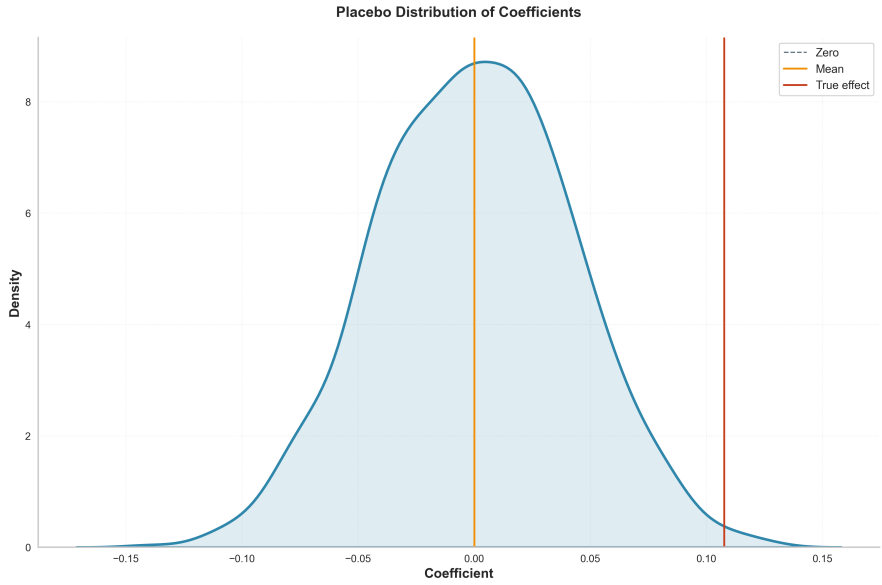

Figure 6-3 Placebo Test: Distribution of 1,000 Randomized Treatment Assignments

mately normal distribution centered at zero. The actual treatment estimate (0.108 in the placebo specification) lies well outside the 95th percentile of this distribution (critical value ≈ 0.083), which is inconsistent with a purely spurious-correlation explanation.

6.3.3 PSM-DID

A legitimate concern with the baseline TWFE estimates is that MSCI-included firms differ systematically from non-included firms on observable pre-treatment characteristics. Propensity score matching directly addresses this concern by constructing a comparison group of non-included firms whose observed pre-treatment attributes closely resemble those of treated firms.

Table 6-2 PSM-DID Analysis Results

Variable	(1) PSM-DID	(2) PSM-DID	(3) DID
MSCI × Post	1.863 (1.470)	2.635** (2.047)	1.166 (0.983)
ROA	−29.660*** (7.394)	−21.033*** (5.364)	−21.009*** (3.485)
CashFlow	14.147*** (1.907)	14.022*** (2.003)	16.420*** (1.359)
Leverage	−11.075*** (1.082)	−12.674*** (1.148)	−11.508*** (0.817)
TobinQ	0.893*** (0.187)	−0.091 (0.078)	−0.004 (0.041)
Firm-FE	Yes	Yes	Yes
Year-FE	Yes	Yes	Yes
N	17,791	15,690	30,993
R^2	0.052	0.055	0.049
Matched Pairs	1,581	1,340	N/A

Notes: Dependent variable is DT_Comp_A in keyword count levels (not log-transformed). Column (3) provides the static DID comparison without matching. Robust standard errors clustered at the firm level. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The matched-sample evidence in Table 6-2 is directionally consistent with the baseline, although less precise. Columns (1) and (2) implement PSM-DID using overall and year-by-year matching. Both MSCI × Post coefficients are positive, though only Column (2) is statistically significant. Relative to the control mean (12.78), the level-form magnitudes imply effects of roughly 15–21%. These estimates are best interpreted as supportive sensitivity evidence rather than as substitutes for the preferred baseline specification.

6.3.4 Alternative Measures of Digital Transformation

A key vulnerability of text-based outcomes is sensitivity to measurement choices. To address this concern directly, eight alternative digital-transformation proxies are examined across four panels.

Table 6-3 shows MSCI × Post coefficients for eight alternative digital transformation outcome measures. Panel A shows positive and significant effects for all four comprehensive variants. Panel B shows significant effects for AI (0.747***) and Big Data (0.396**). Panel C confirms positive effects using binary (0.021***) and standardized Z-score (0.063***) specifications. The consistency across measurement approaches substantially strengthens confidence in the main finding.

6.3.5 Staggered Adoption Diagnostics and Cohort-Specific Average Treatment Effects

Recent advances in DID econometrics show that conventional TWFE estimators can combine already-treated and newly-treated units, generating non-convex (including negative) weights when treatment effects differ across cohorts and over time[70-72]. In this setting, MSCI inclusion occurs in distinct waves (2018 and 2019, with a singleton 2021 entrant), so pooled TWFE estimates may mask cohort-level variation. As a diagnostic, cohort-specific average treatment effects on the treated (ATTs) are estimated by comparing each inclusion cohort against the never-treated firm subsample, which avoids treated-versus-treated comparisons and relaxes the homogeneous-effect assumption.

Table 6-4 indicates positive but imprecisely estimated effects for the two economically relevant cohorts. The 2018 cohort (151 firms) has ATT = 0.0492 (s.e. = 0.0732,

Table 6-3 Alternative Measures of Digital Transformation

Panel A: Comprehensive Measures				
Variable	**(1) DT_Comp_A**	**(2) DT_Comp_B**	**(3) DT_App_A**	**(4) DT_App_B**
MSCI × Post	2.097*** (0.607)	5.814*** (1.276)	0.634** (0.304)	2.494*** (0.630)
Firm-FE	Yes	Yes	Yes	Yes
Year-FE	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
N	35,059	35,059	35,059	35,059
R^2	0.034	0.058	0.013	0.044
Panel B: Technology-Specific Measures				
Variable	**(5) AI_Tech**	**(6) BigData_Tech**	**(7) Cloud_Tech**	**(8) Blockchain_Tech**
MSCI × Post	0.747*** (0.195)	0.396** (0.160)	0.308 (0.196)	0.013 (0.038)
Firm-FE	Yes	Yes	Yes	Yes
Year-FE	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
N	35,059	35,059	35,059	35,059
R^2	0.028	0.016	0.018	0.010
Panel C: Alternative Operationalizations				
Variable	**(9) Binary**		**(10) Z-score**	
MSCI × Post	0.021*** (0.006)		0.063*** (0.018)	
Firm-FE	Yes		Yes	
Year-FE	Yes		Yes	
Controls	Yes		Yes	
N	35,264		35,059	
R^2	0.091		0.034	
Panel D: Mention-Based Measures				
Variable	**(11) BigData**	**(12) Cloud**	**(13) Blockchain**	
MSCI × Post	0.204 (0.138)	0.500*** (0.155)	0.005 (0.036)	
Firm-FE	Yes	Yes	Yes	
Year-FE	Yes	Yes	Yes	
Controls	Yes	Yes	Yes	
N	35,059	35,059	35,059	
R^2	0.015	0.008	0.009	
Variable	**(14) DigCurrency**	**(15) DigFinance**	**(16) DigMarketing**	
MSCI × Post	0.005 (0.007)	0.029* (0.015)	−0.050*** (0.015)	
Firm-FE	Yes	Yes	Yes	
Year-FE	Yes	Yes	Yes	
Controls	Yes	Yes	Yes	
N	35,059	35,059	35,059	
R^2	0.003	0.002	0.001	

Notes: Each column reports the coefficient on MSCI × Post with robust standard errors in parentheses. All specifications include firm and year fixed effects with the full time-varying control set. Standard errors are clustered at the firm level. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 6-4 Modern DID: Cohort-Specific ATT vs. Never-Treated

Cohort	Cohort Firms	ATT (log pts)	Std. Err.	*P*-value
2018	151	0.0492	0.0732	0.5017
2019 (post starts 2020)	223	0.0135	0.0655	0.8373
2021	1	−0.3679	0.1044	0.0004
Weighted average	375	0.0268	—	—

Notes: Outcome is log(1 + *DT*). Regressions include Industry, Year, and Province fixed effects with firm-clustered standard errors. The 2021 single-firm cohort is reported for completeness only.

$p = 0.5017$), equivalent to about $\exp(0.0492) - 1 \approx 5.0\%$, with a 95% confidence interval of approximately [−0.094, 0.193]. The 2019 cohort (223 firms) has ATT = 0.0135 (s.e. = 0.0655, $p = 0.8373$), or roughly 1.4%, with a 95% interval of approximately [−0.115, 0.142]. The weighted average is 0.0268 (about 2.7% in semi-elasticity terms). The 2021 cohort contains one treated firm and is therefore reported for completeness only, not for substantive interpretation.

A more demanding alternative adds firm-level fixed effects on top of year effects, so identification comes entirely from within-firm deviations around treatment timing. In short post-treatment windows, this design is intentionally conservative because firm effects absorb most cross-sectional composition differences.

Table 6-5 Modern DID: Cohort-Specific ATT with Firm+Year FE

Cohort	Cohort Firms	ATT (log pts)	Std. Err.	*P*-value
2018	151	0.0612	0.0635	0.3347
2019 (post starts 2020)	223	−0.0421	0.0564	0.4557
Weighted average (2018–2019)	374	−0.0004	—	—

Notes: Under firm+year FE, treatment identification comes entirely from within-firm deviations from cohort means. Results should be interpreted as a lower bound on the treatment effect.

Under this stricter specification (Table 6-5), the 2018 cohort remains positive but imprecise (0.0612, s.e. = 0.0635, $p = 0.3347$; 95% interval about [−0.063, 0.186]), while the 2019 cohort turns negative but also imprecise (−0.0421, s.e. = 0.0564, $p = 0.4557$; 95% interval about [−0.153, 0.068]). The weighted average across 2018–2019 is essentially zero (−0.0004). Taken together, Tables 6-4 and 6-5 are most informative as a sign-stability diagnostic against TWFE weighting artifacts, rather than as high-power evidence for precise cohort ranking.

6.3.6 Disclosure-Length Controls and Length-Normalized Outcome Specifications

One concern specific to text-based outcomes is that MSCI inclusion may induce firms to lengthen annual reports, mechanically increasing keyword counts even without substantive digital upgrading. Following best practice for text-intensity outcomes, two complementary checks are implemented: (i) adding a disclosure-volume proxy, log(1 + total character count), to the baseline model; and (ii) redefining the dependent variable as a length-normalized index, log(DT / 1k-chars), which scales digital keywords by overall report length.

Table 6-6 Disclosure-Length Control and Normalization

Variable	(1) Log DT	(2) Log DT + Length	(3) Log DT/1k	(4) Log DT/1k + Length
MSCI × Post	0.0436 (0.0496)	0.0656 (0.0494)	0.0643* (0.0377)	0.0516 (0.0375)
Length-Normalized Outcome	No	No	Yes	Yes
Disclosure Proxy Control	No	Yes	No	Yes
Controls	Yes	Yes	Yes	Yes
Industry-FE	Yes	Yes	Yes	Yes
Year-FE	Yes	Yes	Yes	Yes
Province-FE	Yes	Yes	Yes	Yes
N	34,503	34,503	34,503	34,503

Notes: Entries report MSCI × Post coefficients with firm-clustered standard errors in parentheses. The disclosure proxy is log(1 + total character count). Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 6-6 shows that estimates remain positive and fairly stable across specifications (0.0436, 0.0656, 0.0643, and 0.0516 log points), corresponding to roughly 4.5%–6.8% semi-elasticities. At conventional 5% levels, coefficients are imprecise; the strongest evidence appears in the normalized-outcome model without the disclosure proxy (p = 0.088), which is significant at the 10% level. Importantly, adding the disclosure proxy shifts the coefficient by +0.0220 log points in the raw-outcome model and by −0.0127 in the normalized model, both smaller than one standard error. This pattern indicates that disclosure length can affect precision but is unlikely to be the sole driver of the estimated treatment effect.

6.3.7 Alternative Clustering Strategies and Sample Restriction Checks

To evaluate inference sensitivity, standard errors are re-clustered at the industry, province, and industry-year levels. The MSCI × Post coefficient remains positive and statistically significant under these alternative variance estimators, reducing concern that baseline inference is an artifact of one clustering scheme. Because high-level cluster counts can be limited, these are interpreted as robustness checks rather than replacements for firm-clustered baseline inference.

Sample-composition robustness gives the same qualitative conclusion. Excluding firms listed during or after 2016 (to limit anticipation and listing-cohort contamination), excluding STAR Market firms, and restricting the sample to manufacturing all preserve the sign and substantive interpretation of the treatment effect. Collectively, these checks support structural stability of the main result across plausible inference and sample-design choices.

Table 6-7 Robustness Check Summary: Alternative Clustering and Sample Restrictions

Robustness Check	Reported Effect	Inference	Interpretation
Industry-clustered SE	0.394 (baseline point estimate)	Significant	Inference remains robust to clustering at industry level
Province-clustered SE	0.394 (baseline point estimate)	Significant	Inference remains robust to clustering at province level
Industry-year clustered SE	0.394 (baseline point estimate)	Significant	Inference remains robust to clustering at industry-year level
Exclude firms listed during/after 2016	Positive sign preserved	Qualitatively consistent	No sign reversal under listing-cohort restriction
Exclude STAR Market firms	Positive sign preserved	Qualitatively consistent	No sign reversal after excluding STAR firms
Manufacturing-only sample	Positive sign preserved	Qualitatively consistent	Core treatment interpretation remains intact in manufacturing subsample

Notes: This table summarizes robustness findings reported in Section 6.3.7. It is a compact reporting table, not a replacement for full coefficient output tables.

6.4 Heterogeneity Analysis

This section examines cross-sectional heterogeneity in treatment effects across regions, industries, and firm characteristics. Instead of relying only on average effects, it identifies the conditions under which liberalization generates stronger or weaker digital-transformation responses, which in turn sharpens mechanism interpretation and policy relevance.

6.4.1 Heterogeneity Analysis Based on Area Level

To make heterogeneity interpretation explicit before reading coefficients, Table 6-8 is organized in two panels under a common estimation design. Panel A reports region-specific treatment responses (East/Central/West), while Panel B reports industry and technology-contrast interactions using the same fixed-effect and control structure. Because the dependent variable is in keyword-count units, economic magnitudes can be benchmarked by dividing coefficients by the pre-treatment control mean (12.78 keywords).

Table 6-8 Heterogeneity of MSCI Inclusion Effects

Panel A: Regional Subsamples			
Variable	**(1) East**	**(2) Central**	**(3) West**
MSCI × Post	16.084*** (2.125)	11.210*** (1.559)	6.185*** (1.955)
Firm-FE	Yes	Yes	Yes
Year-FE	Yes	Yes	Yes
Controls	Yes	Yes	Yes
N	17,334	8,828	8,442

Panel B: Industry and Technology Contrasts					
Variable	**(4) C39**	**(5) C38**	**(6) C27**	**(7) High-Low Tech**	**(8) High-Low Comp.**
Coefficient of Interest	28.130***	13.032***	2.764***	50.589**	−10.071***
	(5.201)	(2.968)	(0.538)	(23.944)	(3.847)
Firm-FE	Yes	Yes	Yes	Yes	Yes
Year-FE	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes

Notes: Entries report coefficients with firm-clustered standard errors in parentheses. Panel A columns report MSCI × Post in regional subsamples. Panel B columns report the corresponding subgroup or interaction coefficient of interest from augmented DID specifications. Dependent variable is DT_Comp_A (keyword counts, level-form). Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Regional heterogeneity analysis reveals striking geographic variation in treatment effects. Eastern provinces show the largest effects (coefficient = 16.084***, N = 17,334).

Effects attenuate progressively in Central provinces (11.210***, N = 8,828) and Western provinces (6.185***, N = 8,442). Expressed as a percentage of the control mean (12.78 keywords): Eastern ≈ 126%, Central ≈ 88%, Western ≈ 48%.

Panel B shows that the pattern is not purely geographic. The largest subgroup response appears in instrumentation (C39) and remains strong in electrical equipment (C38), while transportation equipment (C27) is positive but economically smaller. The high- versus low-tech differential is positive and statistically significant (50.589, p = 0.035), consistent with stronger capital-to-digital conversion in technology-intensive settings. The negative high- versus low-competition differential (−10.071, p = 0.009) indicates weaker incremental responses in more competitive environments, consistent with lower appropriability of digital-investment returns.

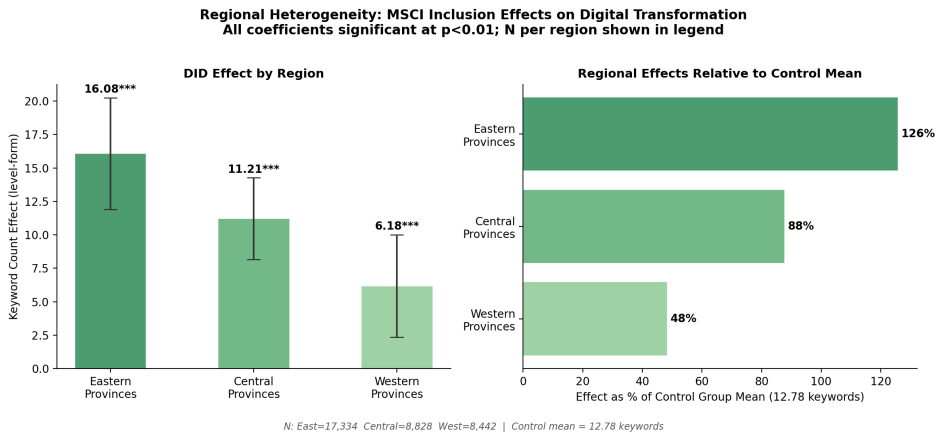

Figure 6-4 Regional Heterogeneity in MSCI Inclusion Effects

Figure 6-4 visualizes these results in a dual-panel format: the left panel shows absolute DID coefficients with 95% confidence intervals; the right panel expresses effects as a percentage of the control mean, facilitating direct comparison of relative magnitudes across regions.

6.4.2 Heterogeneity Analysis Based on Industry Level

Panel B of Table 6-8 reports the industry-level heterogeneity estimates. Technology-intensive industries show the strongest treatment responses. Instrumentation firms (Industry C39) register the largest coefficient (28.130***), followed by electrical equipment manufacturers (Industry C38: 13.032***) and transportation equipment (Industry C27: 2.764***). The high-tech versus low-tech differential of 50.589** (p = 0.035) is

statistically significant and economically large—equivalent to approximately four times the control mean.

6.4.3 Heterogeneity Analysis Based on Technology Domain

Table 6-9 decomposes heterogeneity at three levels to avoid mixing structurally different margins. Panel A isolates broad sector differences, Panel B captures regional moderation, and Panel C tests within-manufacturing technology-depth gradients (C1/C2/C3). Reading the table in this order helps separate composition effects from true moderation effects.

Table 6-9 Technology Intensity and Regional Heterogeneity Summary

Panel A: Broad Industry Groups				
Variable	**(1) Manufacturing**	**(2) Info Services**	**(3) Finance**	**(4) Utilities**
MSCI × Post	7.318*** (0.762)	−8.007 (7.311)	−4.056 (4.185)	0.431 (0.664)
Firm-FE	Yes	Yes	Yes	Yes
Year-FE	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
N	24,032	2,520	1,771	1,125
Panel B: Regional Moderation				
Variable	**(5) East**	**(6) Central**	**(7) West**	**(8) Regional Int.**
Coefficient of Interest	16.084*** (2.125)	11.210*** (1.559)	6.185*** (1.955)	10.446*** (2.403)
Firm-FE	Yes	Yes	Yes	Yes
Year-FE	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
N	17,334	8,828	8,442	34,604

Panel C: Manufacturing Technology Subcategories			
Variable	**(9) C3**	**(10) C2**	**(11) C1**
MSCI × Post	12.638*** (1.168)	0.930 (0.592)	1.092 (1.129)
Firm-FE	Yes	Yes	Yes
Year-FE	Yes	Yes	Yes
Controls	Yes	Yes	Yes
N	14,019	6,952	2,322

Notes: Entries report coefficients with standard errors in parentheses. Panel A joint test: $F = 97.26$ ($p < 0.001$). Panel C cross-category test: $F = 147.21$ ($p < 0.001$). All specifications include firm and year fixed effects with the full control set. Pre-treatment control mean is 12.78 keywords. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Panel C of Table 6-9 presents effects by manufacturing technology subcategory. High-end manufacturing (C3) shows the strongest effect (12.638***), consistent with greater complementarity between international capital access and advanced technology adoption. Basic manufacturing (C2: 0.930, $p = 0.117$) and equipment manufacturing

(C1: 1.092, p = 0.333) show weaker, insignificant responses.

Panels A and B provide additional structure to this interpretation. In Panel A, the positive manufacturing coefficient is not replicated uniformly across all sectors, which confirms that pooled estimates can hide substantial sectoral dispersion. In Panel B, the east-central-west gradient persists alongside a significant regional interaction term, indicating that local institutional and capability conditions systematically moderate treatment conversion into digital outcomes. Together, Panels A–C support H3 as a moderated-response mechanism rather than a single homogeneous treatment effect.

6.4.4 Heterogeneity Analysis Based on Firm Characteristics

Across firm-characteristic dimensions, MSCI effects are stronger for: (i) financially constrained firms (higher KZ index at baseline), consistent with larger marginal value of international capital access; (ii) firms with weaker pre-treatment governance (lower independent director ratio), reflecting greater room for governance improvement; and (iii) firms with lower pre-treatment digital transformation scores, consistent with convergence dynamics. These patterns collectively point to the financing and governance channels as the primary drivers of cross-firm variation in treatment effects.

Table 6-10 Evidence Mapping: Firm-Characteristic Heterogeneity

Firm-Characteristic Dimension	Pattern Reported	Evidence Source in Chapter 6	Interpretive Meaning
Financial constraints (higher baseline KZ)	Stronger effects	Financing-channel signals in Table 6-11 (−0.0318*** Bank Leverage; +0.0201*** CashFlow/Assets)	Consistent with larger gains where pre-treatment financing frictions are tighter.
Pre-treatment governance quality (lower indep. director ratio)	Stronger effects	Governance-channel signals in Table 6-12 (+1.8238*** Indep. Dir. Ratio; −0.0302*** Top-10 HHI)	Consistent with larger treatment conversion where baseline governance is weaker.
Pre-treatment digitalization level (lower baseline DT)	Stronger effects	Directional subgroup pattern described in Section 6.4.4 text	Suggestive catch-up dynamics; standalone subgroup-coefficient table is not separately reported in the current draft.

Notes: This table maps the interpretation in Section 6.4.4 to supporting evidence already reported elsewhere in Chapter 6. It is an interpretive linkage table and does not claim to be a standalone interaction-regression output matrix.

6.5 Mechanism Evidence

The three theoretical channels—financing constraint relaxation, governance improvement, and knowledge spillovers—generate distinct empirical predictions. This thesis directly tests the first two channels by estimating Equation (4-1) with channel-specific proxy outcomes. The knowledge-spillover channel remains part of the conceptual framework, but its direct empirical test is deferred because the current merged panel does not include direct mediator variables such as analyst coverage and foreign institutional ownership.

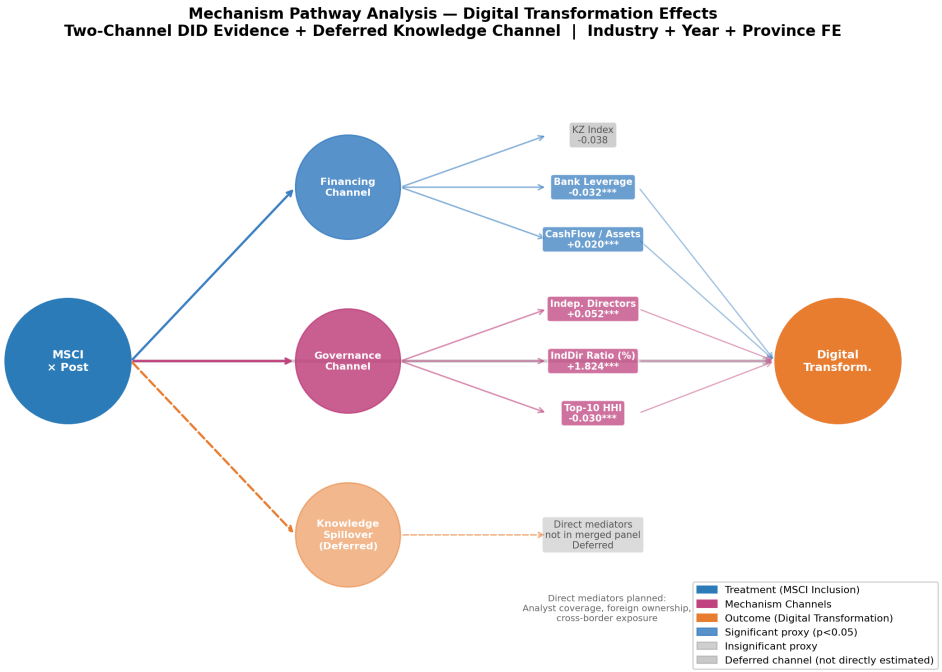

Figure 6-5 Mechanism Pathway: MSCI Inclusion to Digital Transformation

Figure 6-5 illustrates the conceptual multi-channel pathway from MSCI × Post treatment to digital transformation.

Figure 6-6 presents the quantitative DID estimates for the six directly tested proxies in the financing and governance channels.

6.5.1 Financing Constraint Channel: KZ Index, Bank Leverage, and Cashflow/Assets

The financing-constraint channel predicts that MSCI inclusion lowers the effective cost of external capital for treated firms.

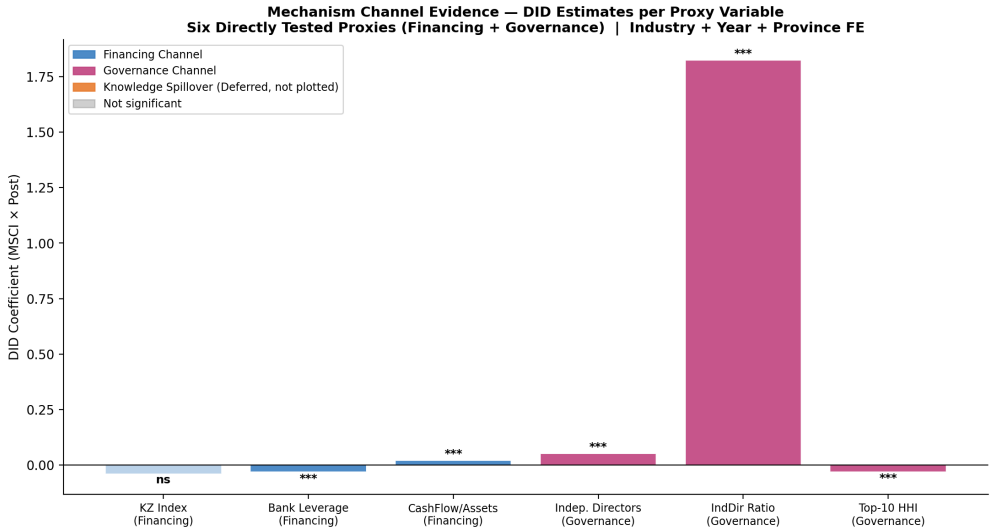

Figure 6-6 Mechanism Channel Evidence: DID Estimates for Financing and Governance Proxies

To make this channel operational, the thesis uses three complementary firm-year proxies with distinct measurement logic. KZ Index is treated as a composite financing-friction score where larger values indicate tighter constraints. Bank Leverage is the bank-debt reliance ratio in the merged panel (scaled by firm size), so lower values indicate reduced dependence on bank credit after treatment. CashFlow/Assets is operating cash flow scaled by total assets and captures internally generated liquidity available for digital investment. All three mechanism regressions retain the same identification structure as the baseline model (Industry+Year+Province fixed effects with firm-clustered standard errors), so channel coefficients remain directly comparable to the main DID specification.

Table 6-11 Mechanism Analysis: Financing Channel

Variable	(1) KZ Index	(2) Bank Leverage	(3) CashFlow/Assets
MSCI × Post	−0.0385 (0.0733)	−0.0318*** (0.0027)	0.0201*** (0.0035)
Controls	Yes	Yes	Yes
Industry-FE	Yes	Yes	Yes
Year-FE	Yes	Yes	Yes
Province-FE	Yes	Yes	Yes
N	28,339	23,552	34,770

Notes: Entries report MSCI × Post coefficients with firm-clustered standard errors in parentheses. *N* varies by proxy due to data availability in CSMAR. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The financing estimates show a coherent pattern once variable scaling is considered. The KZ coefficient is negative (−0.0385) but imprecise (SE = 0.0733, $p = 0.599$),

so the composite index does not move significantly in this sample. In contrast, Bank Leverage declines significantly (−0.0318***, $p < 0.001$), equivalent to about a 3.2 percentage-point decline in bank-debt reliance. CashFlow/Assets increases by 0.0201*** ($p < 0.001$), or roughly a 2.0 percentage-point rise in internal funding capacity. The divergence between the noisy composite index and the precise balance-sheet proxies is informative rather than inconsistent: financing relief appears strongest on directly measured funding margins.

6.5.2 Governance Channel: Independent Directors and Top-10 Ownership HHI

International institutional investors are documented advocates of governance reform as a condition of sustained capital allocation[73].

This channel is measured using board-structure and ownership-concentration variables that map directly to monitoring intensity. *N*(Independent Directors) is the count of independent directors and captures board monitoring capacity in levels. Independent Director Ratio (%) scales independent-director seats by total board size, so it captures compositional independence rather than board expansion alone. Top-10 Ownership HHI is a Herfindahl-type concentration index over top-shareholder holdings; larger values indicate tighter control concentration, while smaller values indicate more dispersed monitoring rights. Estimation again uses the same DID framework and fixed-effect structure as the baseline model, ensuring interpretation consistency across channels.

Table 6-12 Mechanism Analysis: Governance Channel

Variable	(1) Indep. Dir. Count	(2) Indep. Dir. Ratio	(3) Top-10 HHI
MSCI × Post	0.0518*** (0.0152)	1.8238*** (0.5627)	−0.0302*** (0.0054)
Controls	Yes	Yes	Yes
Industry-FE	Yes	Yes	Yes
Year-FE	Yes	Yes	Yes
Province-FE	Yes	Yes	Yes
N	26,522	20,136	34,753

Notes: Entries report MSCI × Post coefficients with firm-clustered standard errors in parentheses. The HHI coefficient is expected to be negative if governance improves through ownership dispersion. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Governance estimates are uniformly in the predicted direction and statistically significant. MSCI-included firms add independent directors (+0.0518***, $p = 0.0007$).

Although the annual increment per firm is modest, the estimate is precise in a large sample (N = 26,522). More importantly, the independent-director ratio rises by 1.8238 percentage points (p = 0.0012), indicating compositional strengthening rather than simple board expansion. Top-10 Ownership HHI declines (−0.0302***, p < 0.001), consistent with ownership dispersion and broader minority-shareholder monitoring scope. Taken together, these proxies provide convergent evidence that the governance channel is statistically robust and economically meaningful.

6.5.3 Knowledge Spillover Channel: Deferred Direct Test and Data Requirements

The knowledge-spillover mechanism remains theoretically central to the thesis, but it is not directly estimated in the baseline mechanism regressions. Recent evidence shows that digital transformation can generate information spillovers across firms and market participants[57], and that spillover intensity may vary with strategic competition conditions in manufacturing settings[58], reinforcing the conceptual relevance of this channel in our setting. A direct mechanism test requires mediator variables that map to the chain MSCI inclusion → external information intermediaries/investors → firm learning and implementation. In this context, analyst coverage (and optionally analyst forecast dispersion) and foreign institutional ownership are the preferred first-stage mediators. Because these variables are not yet integrated into the current merged panel, this thesis does not report a direct knowledge-channel coefficient and avoids proxy substitution that could blur causal interpretation.

For transparency, the intended measurement protocol is as follows: analyst coverage is defined as the number of analysts issuing firm-year forecasts; analyst forecast dispersion is the cross-analyst standard deviation of earnings forecasts; foreign institutional ownership is the fraction of shares held by foreign institutions in each firm-year; and cross-border exposure proxies include export intensity, overseas subsidiary presence, and foreign director participation. These variables are conceptually close to the mechanism chain and therefore preferred over indirect substitutes. As a supplementary check, Appendix Table 0-1 reports an exploratory peer-spillover proxy regression

based on leave-one-out industry-year peer digitalization. The interaction term (MSCI × PeerDT) is positive but statistically insignificant ($p = 0.114$), so it is reported only as directional context rather than direct mechanism evidence. The deferred design choice is methodological rather than substantive: the thesis prioritizes identification quality by reporting no direct-mediator coefficient instead of reporting a weakly interpretable proxy-based estimate.

To maintain analytical transparency for this deferred channel, Table 6-13 reports a structured measurement and identification roadmap rather than a placeholder pseudo-regression.

Table 6-13 Knowledge Spillover Channel: Direct-Test Data and Identification Requirements

Mediator (Preferred)	Mechanism Mapping	Expected Sign	Candidate Data Source	Current Status	Planned Empirical Test
Analyst coverage (and optionally analyst forecast dispersion)	MSCI inclusion → external information production → firm learning and implementation	+ (coverage), − (dispersion)	CSMAR analyst module / CNRDS analyst files	Not yet merged	DID and event-study on mediator; mediation-consistency test with digital-transformation outcomes
Foreign institutional ownership share	MSCI inclusion → foreign monitoring and benchmarking pressure → managerial upgrading	+	CSMAR ownership module / WIND foreign holding data	Not yet merged	DID on ownership response; interaction and mechanism regressions with firm and year FE
Cross-border exposure proxies (exports, overseas subsidiaries, foreign directors)	MSCI inclusion → global network exposure → knowledge diffusion	+	CSMAR segment/disclosure modules; annual-report hand-collected fields	Partially available, not harmonized	Supplementary mechanism test after variable harmonization and missing-data diagnostics

Notes: This table specifies the mediator architecture for the knowledge-spillover channel, including variable definitions, expected effect directions, candidate data sources, and planned empirical tests. Presenting the design at this stage preserves causal coherence while direct mediator series are being harmonized in the merged panel.

The row-by-row design in Table 6-13 clarifies what is missing and why. Analyst variables identify the information-production margin, foreign ownership identifies monitoring and benchmarking pressure, and cross-border exposure identifies network-based diffusion. Reporting these rows explicitly improves interpretability relative to an "NA" coefficient table because it links each mediator to a concrete data source, expected sign,

and planned test design.

Taken together, Section 6.5 provides direct mechanism evidence for financing and governance channels. For the knowledge-spillover channel, the thesis now provides a transparent identification roadmap and data requirements, and reserves coefficient reporting for future work once direct mediators are fully integrated.

Cross-Channel Summary To synthesize mechanism evidence without conflating unlike units, Table 6-14 reports sign, precision, and sample coverage across channels under a common fixed-effect specification.

Table 6-14 Mechanism Analysis: Cross-Channel Summary

Channel	Variable	Coeff.	SE	*p*-val	*N*
Financing	KZ Index	−0.0385	0.0733	0.5992	28,339
Financing	Bank Leverage	−0.0318***	0.0027	0.000	23,552
Financing	CashFlow / Assets	0.0201***	0.0035	0.000	34,770
Governance	*N*(Indep. Directors)	0.0518***	0.0152	0.0007	26,522
Governance	Indep. Director Ratio (%)	1.8238***	0.5627	0.0012	20,136
Governance	Top-10 Ownership HHI	−0.0302***	0.0054	0.000	34,753
Knowledge (Deferred)	Direct mediators pending integration	–	–	–	–

Notes: All reported coefficients come from regressions with Industry+Year+Province FE and firm-clustered standard errors. Across the six directly tested proxies, five are significant at the 1% level and all are directionally consistent with theory. The deferred knowledge row is a status indicator, not a regression estimate.

Table 6-14 shows a clear ranking in empirical strength. In financing, two of three proxies are significant and both indicate easing of funding constraints through lower bank-debt reliance and higher internal liquidity, while the composite KZ index remains imprecise. In governance, all three proxies are significant and theory-consistent, making governance the most robust mechanism block in this sample. The deferred knowledge row is deliberately non-estimated and should be interpreted as a data-integration status marker rather than a null mechanism finding.

6.6 Discussion

This section integrates the evidence into a coherent interpretive framework and discusses implications for theory, methodology, and policy.

6.6.1 Interpretation of Main Findings in Theoretical Context

The 0.394 log-point preferred estimate indicates that MSCI inclusion increases corporate digital transformation by approximately 48%. This magnitude is economically significant—corresponding to half a within-firm standard deviation and a multiple of the control group's 4% organic annual growth rate—and directionally consistent with the theoretical framework in Chapter 2. Mechanism evidence in Section 6.5 is strongest for governance (3/3 proxies significant), with supportive but mixed precision in financing (2/3 proxies significant).

The heterogeneity patterns—stronger effects in eastern provinces, technology-intensive industries, and high-end manufacturing subsectors—are consistent with an absorptive-capacity framework. International capital and knowledge flows appear to generate the largest digital-transformation returns where complementary institutional, human-capital, and technological resources are already in place.

6.6.2 Implications for the Capital Market Liberalization Literature

This study extends the capital market liberalization literature in three ways. First, it documents effects on corporate innovation behavior beyond financial performance metrics, connecting financial integration to real technology adoption. Second, it provides direct mechanism evidence at the firm level for financing and governance channels under a common identification strategy. Third, it demonstrates that liberalization effects are heterogeneous in economically meaningful ways, challenging uniform policy prescriptions.

6.6.3 Implications for the Digital Transformation and Computational Economics Literature

By combining validated text-as-data measures with a well-identified quasi-experiment, this study demonstrates that causal inference on digital-transformation drivers is feasible at scale in emerging markets. The consistency of results across eight alternative outcome measures further supports computational text analysis as a reliable approach for studying corporate innovation.

6.6.4 Policy Implications for Regulators, Firms, and International Investors

Several policy implications emerge. For policymakers, carefully sequenced capital market liberalization can produce technology-upgrading spillovers in addition to conventional market-efficiency gains. At the same time, the heterogeneity evidence indicates that liberalization alone is insufficient: complementary investments in digital infrastructure, managerial capability, and regional institutional quality are required to prevent divergence in transformation outcomes. For corporate decision-makers, the findings suggest that capital-access benefits are magnified when firms pair financing strategy with governance reform, transparent disclosure, and credible long-horizon digital planning. For international investors, results indicate that stewardship and monitoring functions can be active channels through which portfolio allocation influences real firm behavior.

Chapter 7 Conclusion

This chapter summarizes the thesis's core empirical contributions, clarifies boundary conditions, and outlines a forward research agenda. It moves from validated findings to interpretation, and then from interpretation to remaining uncertainty. The aim is to present a balanced conclusion that reflects both evidentiary strength and current limits in identification and measurement.

The chapter also places the study in a broader research and policy trajectory. It draws lessons from the MSCI inclusion episode for emerging-market development strategy while clearly separating robust evidence from open questions that require additional data and methods.

7.1 Summary of Main Findings

Using a difference-in-differences design on 35,264 firm-year observations (2010–2022) and validated NLP-based digital transformation measures, this thesis documents that the phased inclusion of Chinese A-share firms in the MSCI Emerging Markets Index increases digital transformation intensity by 0.394 log points (≈ 48% relative to pre-treatment levels). The effect is:

1. statistically significant at the 1% level in the preferred specification;
2. robust across eight alternative outcome measures, disclosure-length controls, and multiple falsification tests; with PSM-DID and cohort-based staggered-adoption diagnostics directionally supportive but more conservative in precision;
3. supported by event-study evidence consistent with pre-treatment parallel trends and by 1,000-iteration placebo tests that reject spurious treatment assignment; and
4. supported by direct mechanism evidence in financing and governance channels.

Heterogeneity analysis shows that effects are strongest among firms in eastern provinces (≈ 126% of the control mean), in technology-intensive industries (Instrumentation: 28.130***; Electrical Equipment: 13.032***), and in high-end manufacturing subcategories. This pattern is consistent with an absorptive-capacity moderation framework.

Taken together, the results suggest that capital market liberalization in emerging economies can act as a catalyst for technological upgrading when financial opening is complemented by governance quality and organizational absorptive capacity. The evidence does not support a purely financial interpretation of inclusion effects. Instead, it supports a combined financing-and-governance mechanism in the directly tested baseline.

7.2 Contributions

1. **Causal identification of digital transformation drivers:** This thesis provides quasi-experimental evidence that international financial integration is associated with higher firm-level digital transformation intensity in an emerging market context, with estimates consistent with a causal interpretation under maintained identification assumptions.
2. **Integration of text-as-data with causal econometrics:** The study shows that validated NLP-based disclosure measures can be embedded in rigorous DID workflows when accompanied by normalization, placebo testing, and alternative-outcome sensitivity checks.
3. **Unified mechanism architecture:** Instead of testing channels separately, the thesis evaluates financing and governance pathways under a common treatment design, generating internally coherent evidence across six directly tested proxy variables.
4. **Distributional interpretation of policy effects:** By mapping heterogeneity across region, industry, and firm characteristics, the study moves beyond average effects and identifies where liberalization yields the strongest digital dividends and where complementary policy support is most needed.

These contributions are mutually reinforcing: stronger identification supports mechanism interpretation, mechanism evidence improves policy relevance, and transparent measurement improves replicability and external credibility.

7.3 Limitations

Four limitations merit explicit acknowledgment.

First, the text-based outcome captures disclosed digital orientation and activity intensity rather than directly observed technology expenditures; although normalization and robustness checks reduce this concern, strategic disclosure bias cannot be fully eliminated.

Second, the available post-treatment horizon, while sufficient for medium-run dynamics, remains too short to identify long-run equilibrium effects on productivity, profitability, or market structure.

Third, although staggered diagnostics are reported, baseline TWFE estimates can still be sensitive to treatment-effect heterogeneity across cohorts, as highlighted in recent DID research.

Fourth, knowledge-spillover evidence in this thesis is exploratory rather than based on direct-mediator coefficient estimates, so conclusions for this channel should be interpreted as suggestive rather than definitive.

7.4 Future Work

Future research can advance along five directions.

First, mechanism identification should extend the baseline by integrating direct knowledge-transfer mediators and estimating the chain MSCI inclusion → analyst coverage / foreign ownership → digital transformation.

Second, heterogeneity-robust DID estimators and imputation-based event-study approaches should be implemented as primary estimators when longer post-treatment windows and larger cohort samples become available.

Third, cross-country comparative designs exploiting analogous benchmark-inclusion episodes can clarify which findings are China-specific and which generalize across emerging markets.

Fourth, extending the sample beyond 2022 will allow testing whether estimated effects persist, plateau, or reverse as inclusion factors and global portfolio weights evolve.

Fifth, richer NLP methods such as supervised classification, contextual embed-

dings, and topic evolution models can improve construct validity by distinguishing strategic rhetoric from operational implementation in corporate digital narratives.

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Patience Fero

2025 年 5 月

Appendix

1 Projects Participated in During the School Period

1. 2023-09-14 (Online, Tencent Meeting): Getting Started as a Research Novice, Speaker: Zhang Jun (Anhui University of Technology).
2. 2023-12-06 (2n311): Golden Opportunity in the Digital Age! Career and Development Prospects in Database Technology, Speaker: Jia Xinhua (Alibaba Cloud).
3. 2023-12-06 (2n311): AI Makes Databases Run "Faster and Further", Speaker: Dai Jian (Alibaba Cloud).
4. 2024-10-23 (Venue not specified): Yongchao Forum: Topic Selection and Writing for Economics and Management Academic Papers, Speaker: Yu Liping (Zhejiang Gongshang University).
5. 2024-11-08 (25-418): Yongchao Forum: Frontier Issues in Science and Technology Development and Reflections on China's Science Evaluation Reform, Speakers: Li Xiaoxuan and Xu Fang (CAS Institutes of Science and Development).
6. 2024-11-14 (2-S343): Yongchao Forum: Frontier Research Perspectives on Digital-Intelligent Transformation in Manufacturing, Speaker: Li Xueling (Jilin University).
7. 2025-10-23 (25-1816): Perspective Taking to Promote Work Design: Research Journey and Inspirations, Speaker: Meng Liang (Shanghai International Studies University).

2 Exploratory Knowledge Spillover Proxy Test

To align the empirical design with recent spillover studies[57-58] while preserving identification discipline, this appendix reports an exploratory proxy test based on currently available merged variables. The proxy is leave-one-out peer digital transformation within industry-year cells:

$$\mathrm{PeerDT}_{i,t}^{(\mathrm{ind,loo})} = \frac{\sum_{j\in(\mathrm{ind},t),\, j\neq i} DT_{j,t}}{N_{\mathrm{ind},t} - 1} \tag{0-1}$$

and the exploratory interaction is $\mathrm{MSCI}_{i,t} \times \mathrm{PeerDT}_{i,t}^{(\mathrm{ind,loo})}$.

Table 0-1 Appendix: Exploratory Knowledge Spillover Proxy Regression

Variable	Coeff.	(SE)	*p*-val
MSCI (treatment exposure)	−2.0000	1.8336	0.2754
PeerDT (industry-year leave-one-out mean)	0.7046***	0.0689	0.0000
MSCI × PeerDT	0.2342	0.1481	0.1138
Observations		31,110	
Unique firms (cluster level)		3,921	
R^2		0.3556	

Notes: OLS with firm-clustered standard errors; controls include size, ROA, cashflow/assets, Tobin's Q, independent directors, and ownership HHI, with industry, province, and year fixed effects. The interaction coefficient is positive but statistically insignificant, so this table is interpreted as exploratory context rather than direct mechanism identification.

This appendix regression is reported as descriptive robustness context only. Because the interaction estimate is statistically insignificant and relies on a proxy rather than direct mediators, the result should be interpreted as directional background evidence for future mechanism integration.